COURSE BOOK B.TECH. III YEAR
(Autonomous)

Information Technology (IT)

KIET GROUP OF INSTITUTIONS

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Delhi-NCR, Ghaziabad

Connecting Life with Learning

CELEBRATING 25 YEARS OF EXCELLENCE



CURRICULUM STRUCTURE & SYLLABUS

Effective from the Session: 2026-27



VISION

To become a leading institution nationally in the area of professional education, research & innovation for serving the global community.

MISSION

- To impart quality professional education, skills and values through outcome-based innovative teaching learning process in all spheres.
- To undertake collaborative interdisciplinary research as a co-requisite for professional education and simultaneously solve problems faced by society and industry.
- To create an ambience of innovation, entrepreneurship and consultancy for future leaders and innovators.
- To keep faculty members enthusiastic by continuous professional development and positive working environment.

Department of Information Technology

VISION

To achieve excellence in the field of Information Technology and create competent professionals for the benefit of the world community.

MISSION

- To integrate quality education, values and skills in the field of Information Technology through outcome-based teaching and learning process.
- To undertake research for the development of sustainable IT solutions for real world problems of society and industry.
- To foster an intellectual, innovative and entrepreneurial environment to develop IT leaders.
- To create a culture of collaboration and support among teachers and students for sharing of ideas and knowledge.

B.Tech. Information Technology

PROGRAM EDUCATIONAL OBJECTIVE

Graduates will be able to:

- Exhibit leadership and professional skills for the development of society.
- Pursue careers in IT industries, research, and higher education.
- Work as innovator and pursue career in developing sustainable solutions for the benefit of industry and society.

PROGRAM OUTCOMES

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability fori) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Develop strong coding skills for efficient problem-solving.

PSO2: Demonstrate cybersecurity skills and data management practices to protect information

systems and ensure data integrity.

B.Tech (IT) 5th Sem

S No.	Course Category (AICTE)	Course Category (UGC)	BOS	Course Code	Course Name	Type	Academic Learning (AL)			Continuous Internal Examination (CIE)			End Sem Examination (ESE)	Total Marks	Total Credits
							L	T	P	MSE	CA	TOTAL		CIE+ ESE	
1	PC	Major (Core)	CS	CS310L	Theory of Computation	L	3	0	0	60	15	75	75	150	3
2	PC	Major (Core)	CSE	CS311L	Software Engineering and Design Principles	L	3	0	0	60	15	75	75	150	3
3	HS	AEC	ASH	HS203L	Aptitude - 3	L	1	0	0	-	25	25	25	50	1
4	HS	AEC	ASH	HS204L	Soft Skills for Employability - 1	L	2	0	0	-	100	100	-	100	2
Blended															
5	PC	Major (Core)	IT	IT207B	Principles of Quantum Computing	B	2	0	2	60	15	75	75	150	3
6	PC	Major (Core)	IT	IT208B	Computer Vision & Digital Image Processing	B	2	0	2	60	15	75	75	150	3
7	PE	Major (Core)	-	-	Professional Elective – III	B	2	0	4	80	20	100	100	200	4
Lab/Practical															
8	PC	Major (Core)	CSE	CS311P	Software Engineering and Design Principles Lab	P	0	0	2	-	25	25	25	50	1
9	PC	SEC	CS	CS313P	Problem Solving Lab-1	P	0	0	4	-	100	100	-	100	2
10	PW	Summer Internship	CSIT	IT304P	Project Internship	P	0	0	0	-	50	50	-	50	1
Total Hours : 29 hrs.							**15**	**0**	**14**					**1150**	**23**

B.Tech (IT) 6th Sem

S No.	Course Category (AICTE)	Course Category (UGC)	BOS	Course Code	Course Name	Type	Academic Learning (AL)			Continuous Internal Examination (CIE)			End Sem Examination (ESE)	Total Marks	Total Credits
1	PC	Major (Core)	CSIT	IT204L	High Performance Computing	L	3	0	0	60	15	75	75	150	3
2	PC	Major (Core)	CSE	CS315L	Cryptography and Network Security	L	3	0	0	60	15	75	75	150	3
3	HS	AEC	ASH	HS205L	Aptitude - 4	L	1	0	0	-	25	25	25	50	1
4	OE	Multidisciplinary	-	-	Open Elective-I	L	2	0	0	40	10	50	50	100	2
Blended															
5	PE	Major (Core)	IT	IT321B	Software Testing Techniques and Tools	B	3	0	2	80	20	100	100	200	4
9	PE	Major (Core)	-	-	Professional Elective – IV	B	2	0	4	80	20	100	100	200	4
Lab/Practical															
10	HS	AEC	ASH	HS206P	Soft Skills for Employability - 2	P	0	0	2	-	50	50	-	50	1
11	PC	SEC	CS	CS317P	Problem Solving Lab-2	P	0	0	2	-	50	50	-	50	1
12	PW	Research Project / Dissertation	CSIT	IT305P	Minor Project Work	P	0	0	6	-	75	75	75	150	3
Total Hours : 30 hrs.							**14**	**0**	**16**					**1100**	**22**

Professional Electives (PE) – Offered to IT Students

Course Type (PE)	Artificial Intelligence and Machine Learning	Cloud Operations (Powered by AWS)	Cloud Development (Powered by AWS)	Applied Apple Technologies: Swift, XCode & iOS Development *(Powered by Apple)*
BOS	**IT**	**IT**	**IT**	**CSE**
PE-III (5thSem)	ML Essentials using Python (IT206E)	DevOps Engineering (IT317E)	Cloud Architecture Foundation (IT319E)	iOS App Development & Swift Programming (CS322E)
PE-IV (6thSem)	Agentic AI Foundation and Applications (IT316E)	Cloud Security Principles & Practices (IT312E)	Advanced Cloud Architecture (IT320E)	Advanced iOS Frameworks & App Store Deployment (CS323E)

Open Electives -Offered to IT Students

BOS	ASH	ECE	EEE	ME	KSOM
Open Elective-I (6th Sem)	Reliability Theory & Maintenance (MA202O)	Introduction to Robotic Systems (EC237O)	AIoT for Smart Energy (EE214O)	Advanced Engineering Materials (ME229O)	Business Strategy (MS203O)

Theory Courses Detail Syllabus – V Sem

Course Code: CS310L	Course Name: Theory of Computation	L	T	P	C
		3	0	0	3

Pre-requisite: Discrete Structures & Theory of Logic

Course Objectives:

1. To understand the concept of abstract computational machines.
2. To understand the formal languages and grammars.
3. To understand the relation between the formal languages, grammars, and computational machines.

Course Outcome: After completion of the course, the student will be able to

1. Understand basic concepts of automata theory and formal languages.
2. Construct finite automata for regular expressions and regular languages.
3. Illustrate regular and context-free grammar for formal languages.
4. Construct the pushdown automata for context-free languages.
5. Explore Turing machines for formal languages.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	-	-	-	-	-	-	-	-	-	1
CO2	3	2	2	-	-	-	-	-	-	-	1
CO3	3	2	1	-	-	-	-	-	-	-	1
CO4	3	2	2	-	-	-	-	-	-	-	1
CO5	3	3	2	-	-	-	-	-	-	-	1

Unit 1	Finite Automata	09 hours

Basic Concepts and Automata Theory: Introduction to Theory of Computation- Automata, Computability and Complexity, Alphabet, Symbol, String, Formal Languages, Deterministic Finite Automaton (DFA)- Definition, Representation, Acceptability of a String and Language, Non-Deterministic Finite Automaton (NFA), Equivalence of DFA and NFA, NFA with ε-Transition, Equivalence of NFA's with and without ε-Transition, Minimization of Finite Automata, Myhill-Nerode Theorem.

Unit 2	Finite Automata and Language	09 hours

FA with output and Regular Expressions & Languages**:** Finite Automata with output- Moore Machine, Mealy Machine, Equivalence of Moore and Mealy Machine, Regular Expressions, Transition Graph, Kleene's Theorem, Finite Automata and Regular Expression- Arden's theorem, Algebraic Method Using Arden's Theorem, Regular and Non-Regular Languages- Closure properties of Regular Languages, Pigeonhole Principle, Pumping Lemma, Application of Pumping Lemma, Decidability- Decision properties.

Unit 3	Formal Languages	09 hours

Regular and Non-Regular Grammars: Regular Grammars-Right Linear and Left Linear grammars, Conversion of FA into Regular grammar, Context-Free Grammar (CFG)-Definition, Derivations, Languages, Derivation Trees and Ambiguity, Simplification of CFG, Normal Forms- Chomsky Normal Form (CNF), Greibach Normal Form (GNF), Chomsky Hierarchy.

Unit 4	Push Down Automata	09 hours

Push Down Automata and Properties of Context-Free Languages: Nondeterministic Pushdown Automata (NPDA)- Definition, Moves, A Language Accepted by NPDA, Deterministic Pushdown Automata (DPDA) and Deterministic Context free Languages (DCFL), Pushdown Automata for Context Free Languages, Context Free grammars for Pushdown Automata, Two stack Pushdown Automata, Pumping Lemma for CFL, Closure properties of CFL, Decision Problems of CFL, CYK membership algorithm.

Unit 5	Turing Machine	09 hours

Turing Machines and Recursive Function Theory: Basic Turing Machine Model, Representation of Turing Machines, Language Acceptability of Turing Machines, Techniques for Turing Machine Construction, Modifications of Turing Machine, Turing Machine as Computer of Integer Functions, Universal Turing machine, Linear Bounded Automata, Church's Thesis, Recursive and Recursively Enumerable language, Halting Problem, Post's Correspondence Problem.

Total Lecture hours	45 hours

Textbook

1. Introduction to Automata Theory, Languages and Computation, J.E.Hopcraft, R.Motwani, and Ullman. 3rd edition, Pearson Education Asia.

2. Peter Linz, "An Introduction to Formal Language and Automata", Narosa Publishing house.
3. K.L.P. Mishra and N.Chandrasekaran, "Theory of Computer Science : Automata, Languages and Computation", PHI
4. Introduction to languages and the theory of computation, J Martin, 3rd Edition, Tata McGraw Hill.K.Krithivasan and R.Rama; Introduction to Formal Languages, Automata Theory and Computation; Pearson Education, 2009.

Reference Books

1. Daniel I.A. Cohen, John Wiley, "Introduction to languages and the Theory of Computation".
2. Elements and Theory of Computation, C Papadimitrou and C. L. Lewis, PHI 4. Mathematical Foundation of Computer Science, Y.N.Singh, New Age International.
3. Introduction to the Theory of Computation, Michel Sipser, 2nd Edition, Cengage Learning.

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
60	15	75	150
75			

Course Code: CS311L	Course Name: Software Engineering and Design Principles	L	T	P	C
		3	0	0	3

Pre-requisite: Basic programming knowledge

Course Objectives:

1. To introduce students to software engineering principles and software development life cycle models.
2. To develop understanding of software requirements engineering and requirement modeling techniques.
3. To familiarize students with software design principles, UML modeling, and architectural design strategies.
4. To provide knowledge of software metrics, quality assurance, and software testing techniques.
5. To enable students to design maintainable, scalable, and quality software systems using modern engineering practices.

Course Outcome: After completion of the course, the student will be able to

1. Understand software engineering concepts, SDLC models, and software processes.
2. Analyze and document software requirements using standard techniques.
3. Apply software design principles, UML modeling, and architectural strategies.
4. Evaluate software quality using metrics, testing, and quality assurance techniques.
5. Design software systems using modern design approaches and testing methodologies.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	1	-	1	-	-	-	2
CO2	2	3	2	2	2	-	-	1	2	-	1
CO3	2	2	3	2	3	-	-	1	2	1	1
CO4	2	3	1	3	3	-	1	-	2	-	1
CO5	3	2	3	2	3	1	1	2	2	2	1

Unit 1	Introduction to Software Engineering	09 hours

Introduction To Software Engineering and Its Evolution. Software Crisis and The Need for Systematic Software Development. Characteristics Of Good Software. Overview Of Software Development Life Cycle (SDLC). Comparison Of Software Engineering with Traditional Software Engineering. Software Process Models: Waterfall, Incremental, Spiral, And Agile Models. Role Of Documentation and Standards in Software Engineering. Introduction To CASE Tools. Professional Ethics and Responsibility of Software Engineers.

Unit 2	Software Requirements Engineering	09 hours

Concept Of Software Requirements and Their Importance. Functional And Non-Functional Requirements. Requirement Elicitation Techniques: Interviews, Questionnaires, Observation, Prototyping, FAST And QFD. Requirement Analysis and Specification. Software Requirement Specification (SRS): Structure and Characteristics. Requirement Validation and Management. Use Cases and User Stories. Introduction To Requirement Modeling Using UML Use Case Diagrams.

Unit 3	Software Design Concepts and UML Modeling	09 hours

Basic Concepts of Software Design, Architectural Design, Design Strategies: Function-Oriented Design, Object-Oriented

Design, Top-Down Design, Bottom-Up Design. Low-Level Design Concepts, Modularization, Design Structure Charts, Pseudocode and Flowcharts, Coupling and Cohesion (Types and Measures), UML Modeling: Class Diagram, Sequence Diagram, Activity Diagram, State Diagram.

Unit 4	**Software Metrics, Quality Assurance & Complexity Measures**	**09 hours**

Software Metrics and Measurement, Size-Oriented Metrics, Halstead's Software Science, Function Point (FP) Based Metrics Cyclomatic Complexity, Control Flow Graphs (CFG), Software Quality Assurance (SQA), Verification and Validation, SQA Plans, Software Quality Frameworks: ISO 9001, CMMI, Six Sigma.

Unit 5	**Software Testing & Maintenance**	**09 hours**

Software Testing Fundamentals: Objectives of Testing, Testing Principles, Error, Fault, and Failure Concepts. Levels of Testing: Unit Testing, Integration Testing, System Testing, Acceptance Testing, Regression Testing. Testing Techniques: Functional Testing (Black Box Testing), Structural Testing (White Box Testing), Performance Testing, Load Testing, Stress Testing. Testing Strategies: Top-Down Testing, Bottom-Up Testing, Test Drivers and Test Stubs. Software Maintenance: Need for Maintenance, Types of Maintenance: Corrective, Adaptive, Perfective, Preventive.

Total Lecture Hours	**45 hours**

Textbook:

1. Pressman, R. S., & Maxim, B. R., "Software Engineering: A Practitioner's Approach, 9th Edition, McGraw-Hill Education", (Covers SDLC, requirements engineering, design concepts, and architectural principles.)
2. Sommerville, I., "Software Engineering", 10th Edition, Pearson Education.

Reference Books:

1. Booch, G., Rumbaugh, J., & Jacobson, I. *The Unified Modeling Language User Guide*, 2nd Edition, Addison-Wesley. (Authoritative reference for UML modeling techniques.)
2. Gamma, E., Helm, R., Johnson, R., & Vlissides, J., "*Design Patterns: Elements of Reusable Object-Oriented Software"*, Addison-Wesley. *(Classic reference for design patterns.)*
3. Sommerville, I., "*Software Engineering"*, Pearson Education (earlier editions may also be used as reference).

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	**15**	**75**	**150**
75			

Course Code: HS203L	**Course Name: Aptitude - 3**	**L**	**T**	**P**	**C**
		1	**0**	**0**	**1**

Pre-requisite: NIL

Course Objectives:

1. To gain a strong foundation in basic mathematical concepts, and become proficient in performing calculations, solving equations, and manipulating numerical data.
2. To apply quantitative techniques in engineering, managerial, and real-life problem contexts. Through practice and exposure to different types of questions, students will learn time-saving techniques and develop strategies to effectively manage time during exams or real-world scenarios.
3. To prepare for various competitive exams, such as aptitude tests for job placements, college admissions tests for higher education (GATE/CAT/GRE/GMAT) and government entrance exams. The course will provide the necessary knowledge and practice to perform well in these exams.

Course Outcome: After completion of the course, the student will be able to

1. **understand** fundamental concepts of numbers to solve quantitative aptitude problems.
2. **apply** percentage concepts to accurately calculate and **analyze** profit, loss, discounts, and value changes in real-life and business- Apply concepts of Simple Interest and Compound Interest to analyze and solve quantitative and financial problems effectively in academic, real-life, and competitive exam scenarios.
3. apply the principles of ratio, proportion, and mixtures to **analyze** and solve real-world quantitative problems involving comparisons, allocations, and blending of quantities.
4. apply the principles of Arithmetic's to **analyze** motion related problems, efficiency and work distribution to **evaluate** practical real-life science, engineering and quantitative problems with accuracy.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	1		1							2
CO2	3	1		1							2
CO3	3	1		1							2
CO4	3	1		1							2

Unit 1	Number Systems-I	3 hours

Types of numbers: (Natural, Whole, Integers, Rational, Irrational, Real), Divisibility rules, LCM & HCF (Methods and Applications), Unit digit & last digits, concepts, Assumptions and Simplifications.

Unit 2	Percentages & Applications	3 hours

Basic percentage concepts, Value changes & comparison, Profit, loss & discount, marked price, cost price, selling price relationships, Simple and Compound Interest.

Unit 3	Ratio & Proportion, Averages & Mixtures	3 hours

Ratio & Proportion (Definition of Ratio and Proportion. Type of Proportion (Direct Proportion, Inverse Proportion, Continued Proportion), Definition of Variance (Direct and Inverse), Problem related to age, coins, expenditure and saving, Partnerships). Averages & Mixtures (Weighted averages, Alligation method, Concentration mixing problems)

Unit 4	Time, Speed and Distance (TSD)	3 hours

Basic concepts, definition and terminology related to TSD, Direct and Indirect relation of TSD, Unit Conversions, Average Speed, Relative Speed, Problem on Trains, Problem on Boat & Stream, Race and Games

Unit 5	Time & Work	3 hours

Introduction to the Time and Work, Work and Wages, Problems based on Efficiency, Problems based on Time and work, Problems based on Pipe and Cistern.

	Total Lecture hours	15 hours

Textbook

1. Booklet provided by the Aptitude Team, Training Division -CRPC, KIET.

Reference Books:

1. “Quantitative Aptitude for Competitive Examinations" by R.S. Aggarwal, S. Chand Publication.
2. "How to Prepare for Quantitative Aptitude for the CAT" by Arun Sharma, TMH Publication.
3. "Magical Book on Quicker Maths" by M. Tyra, Published by BSC Publishing Co. Pvt. Ltd
4. "Quantitative Aptitude Quantum CAT" by Sarvesh K. Verma, Arihant Publication.

Reference Websites:

1. https://www.indianbix.com
2. https://www.prepinsta.com
3. https://www.talentsprint.com
4. https://www.careeride.com
5. https://www.smartkeeda.com

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	25	25	50
25			

Course Code: HS204L	Course Name: Soft Skills for Employability - 1	L	T	P	C
		2	0	0	2

Pre-requisite: Students should have basic knowledge of English grammar and should have cleared the Soft Skills Essentials 1 & 2 of B. Tech 2nd year.

Course Objectives:
To enhance the verbal aptitude of the students by strengthening their proficiency in English grammar, vocabulary, and comprehension skills, enabling them to communicate effectively in academic, professional, and competitive examination contexts.

Course Outcome: After completion of the course, the student will be able to:

1. Apply rules of English grammar and sentence structure to construct clear, accurate, and grammatically coherent sentences.
2. Demonstrate enhanced vocabulary usage.
3. Analyze and interpret written passages to answer questions based on reading comprehension, sentence completion, and critical reasoning.
4. Maximize the ability to compose passages solve verbal ability questions commonly asked in placement and recruitment assessments, thereby enhancing overall employability readiness.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	1	3	-	2
CO2	-	-	-	-	-	-	-	1	3	-	2
CO3	-	-	-	-	-	-	-	1	3	-	2
CO4	-	-	-	-	-	-	-	1	3	-	2

Unit 1	About Grammar Skills	8 hours

Tenses and Concord – Concept Revision followed by Practice Exercises on Moodle, Active and Passive voice, Direct & Indirect Narration – Concept Revision followed by Practice Exercises on Moodle, Determiners & Modifiers – Concept Revision followed by Practice Exercises on Moodle, Articles, Conjunctions, Interjections, and Prepositions – Concept Revision followed by Practice Exercises on Moodle

Unit 2	Vocab Building	7 hours

Parallelism, Sentence Improvement & Completion, Conditional Sentences (Practice Exercises on Moodle) Vocabulary – Concept of roots & Affixes, synonyms & antonyms, Idioms & Phrases, Analogies, One-word substitutions, odd-man out, spell check, phrasal verbs (practice exercises on Moodle) Vocabulary.com - Learn Words - English Dictionary

Unit 3	Reading Ability	7 hours

Reading Comprehension & Cloze tests (Practice Exercises on Moodle) https://www.indiabix.com/ Sentence Rearrangement (Jumbled Sentences/Para Jumbles), Missing Sentences (Practice Exercises on Moodle)

Unit 4	Writing Ability	8 hours

Essentials of Writing: Do's and Don'ts of Writing, Writing a Paragraph and Essay Writing (On Word Document checked by Grammarly and display of some sample paragraphs –Poor and good both), Write & Improve with Cambridge

	Total Lecture hours	30 hours

Textbook

1. Lewis, Norman. *Word Power Made Easy*. Goyal Publishers, New Delhi.
2. Wren, P. C., & Martin, H. *High School English Grammar and Composition*. S. Chand Publishing, New Delhi.
3. Sharma, Arun, & Upadhyay, Meenakshi. *How to Prepare for Verbal Ability and Reading Comprehension*. McGraw Hill Education, New Delhi.
4. Aggarwal, R. S., & Aggarwal, Vikas. *Objective General English*. S. Chand Publishing, New Delhi.

References:

1. https://www.indiabix.com/online-test/verbal-ability-test/
2. https://prepinsta.com/
3. https://www.vocabulary.com/
4. https://www.bbc.co.uk/learningenglish

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	100	-	100
100			

Course Code: IT207B	Course Name: Principle of Quantum Computing	L	T	P	C
		2	0	2	3

Pre-requisite: Engineering Mathematics, Calculus, Linear Algebra, Basic Physics, Programming Knowledge

Course Objectives:

The objective of this course is to introduce the foundational concepts of quantum computing in a simplified and application-oriented manner. The course covers classical computation basics, qubits, quantum gates, entanglement, and introductory quantum algorithms. It also provides practical exposure to quantum programming using IBM Quantum and Qiskit.

CO1: Explain the differences between classical and quantum computation, including bits, qubits, superposition, and measurement.

CO2: Apply basic linear algebra concepts to represent quantum states and perform single- and two-qubit gate operations.

CO3: Analyze simple multi-qubit systems and determine outcomes of basic quantum circuits and entangled states.

CO4: Compare classical and quantum approaches for selected problems and explain the working principles of key quantum algorithms such as Deutsch–Jozsa and Grover's algorithm.

CO5: Design and simulate basic quantum circuits using Qiskit or IBM Quantum tools.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	1	2	-	-	-	-	-	2
CO2	3	3	1	2	3	-	-	-	-	-	2
CO3	2	3	1	2	2	-	-	-	-	-	2
CO4	2	3	2	2	2	-	-	1	1	-	2
CO5	2	2	3	2	3	1	1	2	2	1	3

Unit 1 | Classical Computation and Mathematical Foundations | 12 hours

Classical bits and binary representation, Logic gates (AND, OR, NOT, XOR), Universal gates (NAND/NOR concept), Basic combinational circuits (Half Adder and Full Adder – overview), Reversible vs irreversible computation (concept), Introduction to computational complexity (intuitive Big-O notation), Review of linear algebra for quantum computing: vectors, matrices, complex numbers, inner product, normalization, matrix multiplication, unitary matrices (concept).

Unit 2 | Qubits and Single-Qubit Quantum Systems | 12 hours

Introduction to qubits, Dirac notation, Superposition principle, Probability amplitudes and normalization, Measurement in computational basis, Bloch sphere (conceptual understanding), Single-qubit quantum gates (X, Y, Z, H, S, T), Matrix representation of quantum gates, Sequential gate operations, Basic quantum circuit representation, Introduction to quantum simulators

Unit 3 | Multiple Qubits and Entanglement | 12 hours

Tensor product (basic concept), Two-qubit systems, Controlled gates (CNOT), Bell states, Quantum entanglement (concept and examples), Measurement of multi-qubit systems, No-cloning theorem (intuitive explanation), Introduction to decoherence, Basic quantum error types (bit-flip and phase-flip – conceptual understanding).

Unit 4	Introductory Quantum Algorithms	12 hours

Classical vs quantum computational perspective (conceptual comparison), Quantum oracles (basic idea), Deutsch algorithm, Deutsch–Jozsa algorithm, Bernstein–Vazirani algorithm, Grover's search algorithm (intuitive explanation and circuit-level understanding), Brief overview of Quantum Fourier Transform (concept only), High-level overview of Shor's factoring algorithm.

Unit 5	Quantum Communication and Programming	12 hours

Quantum teleportation (concept and circuit), Superdense coding, BB84 quantum key distribution (conceptual understanding), Overview of quantum hardware platforms (ion trap, superconducting qubits – basic idea), IBM Quantum platform, Quantum Composer, Introduction to OpenQASM (basic idea), Introduction to Qiskit, Writing and simulating simple quantum circuits.

Total Lecture Hours	60 hours

Textbook:

1. Quantum Computation and Quantum Information – Michael A. Nielsen & Isaac L. Chuang
2. Introduction to Quantum Computing – Phillip Kaye, Raymond Laflamme & Michele Mosca
3. Introduction to Classical and Quantum Computing – Thomas G. Wong

Reference Books:

1. Quantum Computing – A Gentle Introduction – Eleanor Rieffel & Wolfgang Polak
2. Quantum Computing for Everyone – Chris Bernhardt — MIT Press
3. IBM Quantum / Qiskit Documentation & Tutorials (Online Resources)
4. NPTEL Courses – Introduction to Quantum Computing: Quantum Algorithms and Qiskit, IBM and IITM
5. Research Papers on Quantum Error Correction (IEEE/ACM)

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
60	15	75	150
75			

Course Code: IT208B	Course Name: Computer Vision and Digital Image Processing	L	T	P	C
		2	0	2	3

Pre-requisite: Linear Algebra, Probability, Python Programming

Course Objectives:

The objective of this course is to provide strong fundamentals of digital image processing and computer vision techniques, enabling students to analyze, process, and interpret visual data and build real-world vision-based applications.

Course Outcome: After completion of the course, the student will be able to

1. **Understand** digital image representation and low-level vision concepts.
2. **Apply** image enhancement and frequency domain techniques
3. **Perform** image restoration and morphological processing
4. **Implement** image segmentation and feature extraction methods.
5. **Apply** image compression methods and recognition techniques to design basic computer vision application.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	2	-	-	-	-	-	2
CO2	3	3	2	-	3	-	-	-	-	-	2
CO3	3	3	2	2	3	-	-	-	-	-	2
CO4	3	3	3	2	3	-	-	-	-	-	2
CO5	3	3	3	3	3	-	-	2	2	-	3

Unit 1	Digital Image Fundamentals and Low-Level Vision	12 hours

Introduction to image processing and computer vision, steps in digital image processing, components of image processing systems, elements of visual perception, image sensing and acquisition, image sampling and quantization, pixel relationships and connectivity, color image fundamentals, RGB and HSV/HSI color models, two-dimensional mathematical preliminaries, two-dimensional transforms – Discrete Fourier Transform (DFT) and Discrete Cosine Transform (DCT).
Practical Focus: Image representation and visualization. Color space conversion. Image resizing, rotation, interpolation.

Unit 2	Image Enhancement and Spatial/Frequency Domain Processing	12 hours

Image enhancement techniques, gray-level transformations, histogram processing and equalization, basics of spatial filtering, smoothing and sharpening filters, edge detection operators, introduction to frequency domain processing, Fourier transform properties, smoothing and sharpening frequency domain filters, ideal, Butterworth and Gaussian filters, homomorphic filtering, color image enhancement.
Practical Focus: Noise removal and contrast enhancement. Edge detection using spatial operators. FFT-based image filtering.

Unit 3	Image Restoration and Morphological Processing	12 hours

Image restoration concepts, degradation model and properties, noise models, mean and order statistic filters, adaptive filtering, band-pass, band-reject and notch filters, inverse filtering, Wiener filtering, morphological image processing, erosion and dilation, opening and closing operations, applications of morphological processing.
Practical Focus: Restoration of noisy and blurred images. Comparison of restoration techniques. Morphological processing on real images.

Unit 4	Image Segmentation and Feature Extraction	12 hours

Image segmentation concepts, edge detection and edge linking, Hough transform for line detection, thresholding techniques, region-based segmentation, region growing, region splitting and merging, morphological segmentation, watershed segmentation algorithm, feature extraction concepts, boundary representation, regional descriptors, shape and texture features.
Practical Focus: Object segmentation and contour detection. Feature extraction for shape and texture. Object counting and region analysis.

Unit 5	Image Compression, Recognition and Applications	12 hours

Need for image compression, lossless compression techniques – Run Length Encoding, Huffman coding, arithmetic coding, lossy compression techniques, JPEG and MPEG standards, pattern recognition fundamentals, feature-based recognition, matching techniques, object detection, face detection using Haar cascades, motion detection and background subtraction, introduction to convolutional neural networks for image classification.
Practical Focus: Compression experiments. Face detection system. Motion detection in videos. Basic CNN-based image classifier (introductory).

	Total Lecture Hours	60 hours

Textbook:

1. Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing, 4th Edition, Pearson Education. (Standard AKTU and IIT-level textbook for fundamentals and theory)
2. Anil K. Jain, Fundamentals of Digital Image Processing, Pearson Education. (Clear explanation of image formation and processing concepts)

Reference Books:

1. Richard Szeliski, Computer Vision: Algorithms and Applications, Springer.
   (Widely used in IITs for computer vision algorithms and applications)
2. David A. Forsyth and Jean Ponce, Computer Vision: A Modern Approach, Pearson Education.
   (Comprehensive coverage of vision theory and real-world systems)
3. Milan Sonka, Vaclav Hlavac, and Roger Boyle, Image Processing, Analysis, and Machine Vision, Cengage Learning.
   (Strong on segmentation, morphology, and feature extraction).
4. William K. Pratt, Digital Image Processing, Wiley.
   (Classic reference for advanced image processing techniques).
5. E.R. Davies, Computer and Machine Vision: Theory, Algorithms, Practicalities, Academic Press.
   (Useful for practical vision system design)

Online / Web Resources:

1. NPTEL – Digital Image Processing (IIT Kanpur) (https://nptel.ac.in/courses/117104069)
2. NPTEL – Computer Vision And Image Processing - Fundamentals And Applications (IIT Guwahati) (https://nptel.ac.in/courses/108103174)
3. OpenCV Official Documentation (https://docs.opencv.org/5.x/index.html)
4. PyImageSearch Tutorials (https://pyimagesearch.com)

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	15	**75**	**150**
75			

Professional Elective-III Courses Detail Syllabus – V Sem

Course Code: IT206E	**Course Name: Machine Learning Essentials using Python**	**L**	**T**	**P**	**C**
		2	**0**	**4**	**4**

Pre-requisite: Linear Algebra, Probability & Statistics, Python Programming, Data Structures

Course Objectives:

The objective of this course is to provide strong foundations of machine learning techniques, enabling students to design, implement and evaluate intelligent data-driven systems and develop industry-ready ML applications.

Course Outcome: After completion of the course, the student will be able to

1. Understand machine learning fundamentals, statistical learning concepts and ML workflows.
2. Apply regression, probabilistic and classification techniques for predictive modelling.
3. Implement tree-based, SVM and ensemble learning methods for real-world problems.
4. Apply clustering, dimensionality reduction and unsupervised learning for pattern discovery.
5. Apply neural networks, reinforcement learning and genetic algorithms.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	2	-	-	-	-	-	2
CO2	3	3	2	-	3	-	-	-	-	-	2
CO3	3	3	3	2	3	-	-	-	-	-	2
CO4	3	3	2	3	3	-	-	-	-	-	2
CO5	3	3	3	3	3	-	-	2	2	-	3

Unit 1	Machine Learning Foundations and Statistical Learning	18 hours

Introduction to machine learning, AI vs ML vs DL, types of learning, ML pipeline, supervised vs unsupervised learning, data preprocessing, feature engineering, overfitting and bias-variance, train/validation/test split, evaluation strategies.
Case Study: Customer retention prediction using behavioral analytics.
Practical Focus: Data preprocessing, feature engineering, exploratory data analysis, model pipeline creation, training-validation workflow.

Unit 2	Supervised Learning: Regression and Probabilistic Models	18 hours

Linear regression, regularization (Ridge), logistic regression, Naïve Bayes, Bayesian Belief Networks, probabilistic learning concepts, likelihood estimation, classification metrics and regression metrics.
Case Study: House price prediction and price recommendation system.
Practical Focus: Regression implementation, logistic classification, probabilistic modelling, confusion matrix, ROC analysis, performance evaluation.

Unit 3	Supervised Learning: Trees, SVM and Ensemble Methods	18 hours

Decision trees, entropy, information gain, k-Nearest Neighbors, Support Vector Machines, kernel methods, ensemble learning – bagging, boosting, random forest, stacking, cross-validation and hyperparameter tuning.
Case Study: Credit risk prediction for loan approval systems.
Practical Focus: Decision tree implementation, SVM modelling, ensemble training, hyperparameter tuning, model comparison using validation methods.

Unit 4	Unsupervised Learning and Dimensionality Reduction	18 hours

Clustering methods – k-means, DBSCAN, Gaussian mixture concepts, anomaly detection, dimensionality reduction, PCA, feature extraction and representation learning basics.
Case Study: Customer segmentation for targeted marketing.
Practical Focus: Clustering implementation, PCA visualization, anomaly detection, cluster quality analysis, unsupervised workflow design

Unit 5	Neural Networks, Reinforcement Learning and Genetic Algorithms	18 hours

Perceptron, multilayer neural networks, backpropagation, CNN overview, reinforcement learning fundamentals, MDP, Q-learning, introduction to genetic algorithms, population, fitness, selection, crossover, mutation, optimization using GA.
Case Study: Feature selection using genetic algorithm for healthcare prediction.
Practical Focus: Neural network implementation, simple deep learning workflow, Q-learning simulation, GA-based feature optimization.

	Total Lecture Hours	90 hours

Textbook:

1. Tom M. Mitchell, Machine Learning, McGraw Hill
2. Christopher Bishop, Pattern Recognition and Machine Learning, Springer
3. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, O'Reilly

Reference Books:

1. Kevin Murphy, Machine Learning: A Probabilistic Perspective
2. Richard Sutton & Andrew Barto, Reinforcement Learning: An Introduction
3. Hastie, Tibshirani & Friedman, Elements of Statistical Learning
4. Ian Goodfellow, Deep Learning

Online / Web Resources:

1. NPTEL – Machine Learning (IIT Madras / IIT Kharagpur)
2. Coursera – Machine Learning by Andrew Ng
3. Scikit-learn documentation
4. TensorFlow official tutorials

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: IT317E	Course Name: DevOps Engineering	L	T	P	C
		2	0	4	4

Pre-requisite: AWS Cloud Computing & Operations & AWS Cloud Practitioner

Course Objectives:

The primary objective of this course is to equip learners with DevOps engineering skills required to design, automate, deploy, and operate scalable AWS cloud solutions. The course focuses on Infrastructure as Code, CI/CD pipelines, containerization, DevSecOps practices, and architectural best practices aligned with AWS Solutions Architect – Associate certification.

Course Outcome: By the end of this course, learners will be able to:

1. Design scalable and fault-tolerant cloud architectures using DevOps principles.
2. Implement Infrastructure as Code for automated provisioning.
3. Build and manage CI/CD pipelines for application deployment.
4. Containerize and orchestrate applications in cloud environments.
5. Integrate monitoring, security, and reliability practices into DevOps workflows.

Final Academic Outcome / ESE Target: AWS Certified Solutions Architect – Associate (SAA-C03) by Amazon Web Services

CO-PO Mapping (Scale1:Low, 2:Medium, 3:High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	3	-	-	2	2	2	3
CO2	3	3	2	3	3	-	-	2	2	2	3
CO3	2	2	3	2	3	-	-	3	2	2	2
CO4	3	3	2	2	3	2	3	2	2	2	2
CO5	3	2	3	-	3	-	-	2	2	3	3

Unit1	DevOps Foundations & AWS Architecture Patterns	15 hours

DevOps lifecycle, CI/CD concepts, infrastructure automation mindset, immutable infrastructure, high availability design, scalability patterns, microservices basics, AWS Well-Architected Framework pillars. Introduction to Git workflows and AWS CLI usage.

Case Study: Designing a Highly Available Web Architecture Ready for Automated Deployment

Unit 2	Infrastructure as Code & Environment Automation	18 hours

Infrastructure provisioning using Terraform and CloudFormation, modular template design, state management, drift detection, environment parity (Dev/Test/Prod), IAM roles for automation, S3 backend configuration, automated rollback strategies.

Case Study: Provisioning a Complete Application Stack Using Terraform Modules

Unit 3	Continuous Integration & Continuous Delivery	18 hours

Version control workflows, branching strategies, automated build pipelines, artifact management, CodeBuild, CodePipeline, Jenkins integration, blue-green deployments, rolling deployments, canary releases, deployment rollback mechanisms.

Case Study: Implementing a Blue-Green CI/CD Pipeline for a Production Application

Unit 4	Containerization & Orchestration	18 hours

Docker fundamentals, image creation, container networking, container registry usage, ECS architecture, EKS basics, service scaling, rolling updates, cluster management, container-based microservices deployment.

Case Study: Deploying a Containerized Microservice Application Using ECS

Unit5	DevSecOps, Monitoring & Production Reliability	21 hours

DevSecOps principles, IAM best practices for pipelines, secret management, encryption fundamentals, CloudWatch metrics and logs, centralized logging, application monitoring, automated alerting, cost optimization basics, performance tuning strategies.

Case Study: Integrating Monitoring and Security Controls into a Production Deployment Pipeline

Digital Course: AWS CloudOps Engineer - Introduction to Cloud Operations

Exam Readiness: AWS Certified Solutions Architect – Associate (SAA-C03)
Exam Prep Official Practice Question Set: AWS Certified Solutions Architect – Associate (SAA-C03)
Exam Prep Enhanced Course: Solutions Architect – Associate (SAA-C03) Learning Plan (includes labs)

Total Lecture Hours	90 hours

Textbook:

1. Kavis, M. Architecting the Cloud: Design Decisions for Cloud Computing Service Models.
2. Wittig, A. & Wittig, M. Amazon Web Services in Action (Latest Edition).

Reference Books:

1. Yevick, A. & Yevick, D. Terraform: Up & Running.
2. Beyer et al. Site Reliability Engineering.
3. AWS Well-Architected Framework – Official Whitepapers

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: IT319E	Course Name: Cloud Architecture Foundation	L	T	P	C
		2	0	4	4

Pre-requisite: Networking, OS, Database, Programming Language Knowledge

Course Objectives:

- To introduce the fundamental concepts of cloud computing and cloud architecture principles.
- To understand virtualization, cloud deployment models, and service models
- To design scalable, secure, and cost-effective cloud architectures.
- To explore compute, storage, networking, and security services in cloud platforms.
- To build foundational knowledge required for cloud architect roles and certifications

Course Outcome: By the end of this course, learners will be able to

CO1: Explain cloud computing concepts, deployment models, and service models.
CO2: Design basic cloud architecture using compute, storage, and networking components.
CO3: Apply cloud security principles including IAM, encryption, and network isolation.
CO4: Implement scalable and highly available architectures using load balancing and auto-scaling.
CO5: Analyze cost optimization and monitoring strategies in cloud environments.

CO-PO Mapping(Scale1:Low,2:Medium,3:High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	3	-	-	2	2	2	3
CO2	3	3	3	2	3	-	-	2	2	2	3
CO3	3	3	3	2	3	-	-	2	2	2	3
CO4	3	3	3	2	3	-	-	2	3	2	3
CO5	3	3	3	2	3	-	-	2	2	2	3

Unit1	Introduction to Cloud Computing and Architecture Principles	18 hours

Cloud computing fundamentals, evolution of cloud, characteristics (on-demand self-service, elasticity, resource pooling), deployment models (Public, Private, Hybrid, Multi-cloud), service models (IaaS, PaaS, SaaS), shared responsibility model, global infrastructure concepts (regions, availability zones), basic cloud architecture principles (scalability, reliability, fault tolerance).

Case Study: Designing a basic web application on cloud infrastructure.

Unit 2	Compute and Networking Architecture	18 hours
Virtual machines, containers, serverless computing, instance types, auto-scaling groups, load balancing concepts, VPC fundamentals, subnets, route tables, internet gateways, NAT gateways, DNS concepts, CDN basics. Case Study: Designing a scalable 3-tier web architecture using compute and load balancing.		
Unit 3	**Cloud Storage and Database Services**	**18 hours**
Object storage, block storage, file storage, database services (Relational vs NoSQL), high availability storage design, backup strategies, replication, disaster recovery concepts (RTO, RPO), data lifecycle management. Case Study: Architecting a fault-tolerant database system with backup and recovery.		
Unit 4	**Cloud Security and Identity Management**	**18 hours**
IAM concepts, roles and policies, authentication and authorization, encryption at rest and in transit, network security groups, firewall concepts, monitoring and logging, compliance basics, security best practices. Case Study: Designing secure cloud architecture for an e-commerce platform.		
Unit5	**High Availability, Scalability, Monitoring and Cost Optimization**	**18 hours**
High availability architecture, fault tolerance strategies, auto-scaling policies, monitoring tools, logging services, cost management principles, cloud pricing models, architecture best practices framework, well-architected principles. Case Study: Optimizing cost and performance of a production cloud application. Exam Readiness: Exam Guide for AWS Certified Solutions Architect – Associate (SAA-C03) Exam Prep Official Practice Question Set: AWS Certified Solutions Architect – Associate (SAA-C03) Exam Prep Enhanced Course: AWS Certified Solutions Architect – Associate (SAA-C03)		
	Total Lecture Hours	**90 hours**

Textbook:

1. Wittig, M., & Wittig, A. (2021). Amazon Web Services in Action (2nd ed.). Manning Publications.
2 Richards, M., & Ford, N. (2020). Fundamentals of Software Architecture. O'Reilly Media.
3 Erl, T., Puttini, R., & Mahmood, Z. (2013). Cloud Computing: Concepts, Technology & Architecture. Prentice Hall.

Reference Books:

1. Sbarski, P., Cui, Y., & Nair, A. (2022). Serverless Architectures on AWS (2nd ed.). O'Reilly Media.
2.AWS Documentation – Architecture Center.

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
80	20	**100**	**200**
100			

Course Code: CS322E	Course Name: iOS App Development & Swift Programming	L	T	P	C
		2	0	4	4

Pre-requisite: Foundation of iOS App Development

Course Objectives:

1. To develop proficiency in advanced Swift programming and responsive SwiftUI application development.
2. To build data-driven iOS applications using networking, JSON processing, and local data persistence.

Course Outcome: After completion of the course, the student will be able to

CO1: To Apply advanced Swift programming concepts to develop reusable and efficient code.
CO2: To Apply responsive user interfaces using SwiftUI layouts.
CO3: To Develop multi-screen iOS applications with navigation and media integration.
CO4: To Integrate REST APIs and parse JSON data in SwiftUI applications.
CO5: To Develop and debug SwiftUI applications using local data persistence techniques.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	1	2	-	-	-	-	-	2
CO2	2	2	3		3	-	-	-	-	-	2
CO3	2	2	3		3	-	-	-	-	-	2
CO4	3	3	3	2	3	-	-	-	-	-	2
CO5	2	3	3	2	3	-	-	-	-	-	2

Unit 1	Advanced Swift Language Concepts	18 hours

Generics: Writing generic functions and custom types to eliminate redundant code.
Protocol-Oriented Programming: Creating protocols, applying protocol conformance, and using protocol extensions for reusable code.
Memory & Lifecycle Basics: Understanding Automatic Reference Counting (ARC), avoiding simple retain cycles, and managing SwiftUI app states using ScenePhase.

Unit 2	SwiftUI Layouts & Responsive UI	18 hours

Core Layout Containers: Structuring user interfaces cleanly using VStack, HStack, and ZStack.
Adaptive Design: Utilizing GeometryReader to handle view dimensions dynamically.
Screen Adaptability: Best practices for building layouts that scale across different screen sizes.

Unit 3	SwiftUI Navigation & Media Handling	18 hours

Navigation Components: Implementing view transitions using NavigationStack and NavigationLink.
Tab Interfaces: Creating multi-tab architectures using TabView.
Camera & Photo Library: Requesting user permissions, opening the camera/photo library, and displaying selected images in SwiftUI views.

Unit 4	Networking & JSON Parsing	18 hours

Web API Connections: Using URLSession to fetch remote data from simple REST APIs.
Data Conversion: Working with the Codable protocol to map network responses to Swift models.
JSON Handling: Parsing standard JSON data structures and updating the SwiftUI user interface with the results

Unit 5	Data Persistence & Project	18 hours

UserDefaults: Storing and retrieving small user preferences and application settings.
FileManager: Simple techniques for reading and writing data files to the device storage.
API-Based Mini Project: Developing and debugging a simple, integrated SwiftUI application that fetches network data, displays a dynamic layout, and saves user settings.

	Total Lecture Hours	90 hours

Textbook:

1. SwiftUI Essentials – iOS Edition, Neil Smyth, SwiftUI Essentials – iOS Edition, Payload Media, Latest Edition.
2. iOS Programming: The Big Nerd Ranch Guide, Christian Keur and Aaron Hillegass, iOS Programming: The Big Nerd Ranch Guide, Big Nerd Ranch Guides, Latest Edition.

Reference Books:

1. Swift Programming: The Big Nerd Ranch Guide, Matthew Mathias and John Gallagher, Swift Programming: The Big Nerd Ranch Guide, Big Nerd Ranch Guides, Latest Edition.
2. Develop in Swift: Explorations, Apple Education, Develop in Swift: Explorations, Apple Inc., Latest Edition.

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Practical Courses Detail Syllabus – V Sem

Course Code: CS311P	Course Name: Software Engineering and Design Principles Lab	L	T	P	C
		0	0	2	1

Pre-requisite: Basic Programming Knowledge

Course Objectives:

- To apply software engineering concepts in practical software development scenarios.
- To train students in software requirement analysis and UML-based design.
- To expose students to modern software development workflows and testing practices.
- To develop professional documentation and collaborative development skills.
- To familiarize students with software quality assurance and software testing tools.

Course Outcome: After completion of the course, the student will be able to

1. Apply software engineering principles to analyze software requirements and prepare professional software documentation.
2. Develop UML models and software designs using standard design principles and architectural strategies.
3. Apply software metrics, testing techniques, and quality assurance practices using modern software tools.
4. Demonstrate collaborative software development using version control, agile practices, and effective project presentation skills.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	3	2	2	2	-	-	1	2	1	1
CO2	2	2	3	2	3	-	-	2	3	1	1
CO3	2	3	2	3	3	-	1	1	2	1	2
CO4	1	1	2	-	3	-	1	3	3	2	2

List Of Practical's (Indicative & Not Limited To)

Session 1: Introduction to Software Engineering Tools

- Introduction To Git And Github
- Repository Creation, Version Control Basics
- Software Project Lifecycle In Industry
- Team Formation And Project Domain Selection

Deliverable: GitHub Repository Setup
Industry Mapping: Collaborative Development

Session 2: Problem Identification & Feasibility Analysis

- Identification of Real-World Software Problems
- Problem Statement Writing
- Technical and Operational Feasibility
- Project Scope Definition

Deliverable: Problem Statement Document
Industry Mapping: Product Requirement Analysis

Session 3: Requirements Engineering – Elicitation

- Stakeholder identification
- Requirement gathering techniques:
  - ➢ Interview simulation
  - ➢ Questionnaire design
  - ➢ User persona creation
- User story writing

Deliverable: Requirement Gathering Report
Industry Mapping: Business Analysis and Client Interaction

Session 4: SRS Documentation
- Preparation Of Software Requirement Specification (SRS)
- IEEE SRS Format:
  - Introduction
  - Overall Description
  - Functional Requirements
  - Non-Functional Requirements

Tools: MS Word / Google Docs / Overleaf
Deliverable: Complete SRS Document
Industry Mapping: Formal Documentation Standards

Session 5: UML Use Case Modeling
- Use Case Identification
- Use Case Diagram Creation
- Use Case Description Templates

Tools: Star UML / Draw.Io
Deliverable: Use Case Diagram And Narratives
Industry Mapping: Requirement Modeling

Session 6: UML- Class Diagram Design
- Object-Oriented Analysis
- Class Identification, Attributes, Methods, Relationships
- UML Class Diagram Modeling

Deliverable: Class Diagram
Industry Mapping: Object-Oriented System Design

Session 7: UML- Sequence and Activity Diagram Modeling
- Sequence Diagram Design
- Message Flow Representation
- Workflow Modeling Using Activity Diagram

Deliverable: Sequence And Activity Diagram
Industry Mapping: Interaction And Workflow Modeling

Session 8: Software Design Principles
- Modularization
- Coupling and Cohesion Analysis
- Function-Oriented vs Object-Oriented Design
- Top-Down and Bottom-Up Design

Deliverable: Design Analysis Report
Industry Mapping: Software Design Engineering

Session 9: Software Architecture Design and Design Patterns
- Layered Architecture Design
- Client-Server
- Microservices (Conceptual)

Deliverable: System Architecture Diagram
Industry Mapping: System Architecture Planning

Session 10: Software Metrics & Complexity Analysis
- Size-Oriented Metrics
- Halstead Metrics Calculation
- Function Point Analysis
- Cyclomatic Complexity
- Control Flow Graphs

Deliverables: Metrics Calculation Report
Industry Mapping: Software Quality Measurement

Session 11: Unit & Integration Testing
- Writing Test Cases
- Unit Testing Concepts
- Integration Testing Strategies
- White Box Testing Basics

Tools: JUnit / Manual Testing Templates
Deliverable: Test Case Document
Industry Mapping: Software Testing Practices

Session 12: Functional & Performance Testing
- Black Box Testing
- Functional Testing
- Load Testing
- Stress Testing
- Regression Testing

Deliverable: Testing Report
Industry Mapping: Quality Assurance Engineering

Session 13: Agile Workflow & Project Management
- Agile Methodology
- Sprint Planning
- Backlog Creation
- Scrum Board Demonstration

Tool: Trello / GitHub Projects
Deliverable: Agile Sprint board
Industry Mapping: Agile Industry Workflow (Scrum model)

Session 14: Final Project Design Review
- Review of SRS
- UML Review
- Architecture Validation
- Testing Documentation Review

Deliverable: Final Design & Testing Report
Industry Mapping: Software Review Process

Session 15: Final Project Demonstration and Viva
- Final Project Presentation
- UML Model Explanation
- Architecture Walkthrough
- Demonstration Of Testing Reports
- SRS + Design Document Submission

Deliverable: Final Project Documentation
Industry Mapping: Client Demo and Technical Presentation

Total Hours	**30 hours**

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
-	**25**	**25**	**50**
25			

Course Code: CS313P	Course Name: Problem Solving Lab-1	L	T	P	C
		0	0	4	2

Pre-requisite: NIL

Course Objectives:

1. To familiarize students with the basic and advance of Problem Solving.
2. To learn modern techniques to solve DSA Problems.

Course Outcome: After completion of the lab, the student will be able to

1. Apply fundamental algorithmic thinking
2. Implement major problem-solving patterns.
3. Analyze and evaluate the efficiency of solutions.
4. Design solutions using advanced techniques
5. Integrate multiple techniques to solve complex problems

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2	2	-	-	-	-	-	2
CO2	3	3	2	2	2	-	-	-	-	-	2
CO3	3	3	2	2	2	-	-	-	-	-	3
CO4	3	3	2	2	3	-	-	-	-	-	3
CO5	3	3	2	3	2	-	-	-	-	-	3

List of Experiments:

1. Basic Logic Building - 1. Print all divisors 2. Prime check 3. GCD using Euclid
2. Number Theory Basics - 1. Sieve of Eratosthenes 2. Count primes in range 3. Modular arithmetic problems
3. Recursion-I - 1. Factorial/Power recursion 2. Print subsets 3. Fibonacci recursion
4. Recursion-II/Backtracking Basics - 1. N-Queen (4×4) 2. Rat in a maze 3. Print permutations
5. Two Pointer Technique–I - 1. Pair sum 2. Remove duplicates from sorted array 3. Move zeros to end
6. Two Pointer Technique–II - 1. 3-sum 2. Container with most water 3. String palindrome check
7. Sliding Window–I - 1. Max sum subarray (fixed k) 2. Count anagram occurrences 3. Longest substring without repeat
8. Sliding Window–II - 1. Minimum window substring 2. Longest repeating character replacement 3. Fruit into baskets
9. Binary Search–I - 1. First/last occurrence 2. Square root using BS 3. Search in rotated sorted array
10. Binary Search on Answer - 1. Allocate books 2. Aggressive cows 3. Koko eating bananas
11. Greedy–I - 1. Activity selection 2. Minimum coins 3. Jump game
12. Greedy-II - 1. Job sequencing 2. Gas station problem 3. Minimum platforms
13. Bit Manipulation–I - 1. Count set bits 2. Find unique element using XOR 3. Check power of 2
14. Bit Manipulation–II - 1. Subset generation using bits 2. Swap bits in number 3. Count bits from 1 to N

Lab-1 Mini Project - 1. Pattern-based problem sheet (mix) 2. Contest simulation 3. 10 mixed technique problems

Total Laboratory Hours	60 hours

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	100	-	100
100			

Theory Courses Detail Syllabus – VI Sem

Course Code: IT204L	Course Name: High Performance Computing	L	T	P	C
		3	0	0	3

Pre-requisite: Computer Organization & Architecture, Data Structures

Course Objectives:

This course aims to introduce the fundamental concepts of parallel computing and high-performance systems, along with an understanding of parallel architectures and interconnection networks. It develops practical skills in message passing and shared memory programming and optimize the performance of parallel programs.

Course Outcome: After completion of the course, the student will be able to

CO1: Understand parallel computing models and architectures

CO2: Design and analyze parallel algorithms in comparison with linear algorithms.

CO3: Understanding the performance parameters and optimization techniques for parallel processing.

CO4: Implement parallel algorithms using modern architectures and programming models including shared memory, distributed systems (MPI).

CO5: Implement performance evaluation and optimization techniques and Gain exposure to real-world HPC systems (GPU clusters, DGX systems)

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	-	-	-	-	-	-	-	2
CO2	3	3	2	1	-	-	-	-	-	-	2
CO3	2	2	2	2	2	-	-	1	-	-	2
CO4	2	2	2	2	2	-	-	1	-	-	2
CO5	2	2	-	2	3	-	-	1	-	-	-

Unit 1	Introduction to High Performance Computing	15 hours

Era of Computing: Evolution from serial to parallel computing, Need and applications of High Performance Computing, Sequential vs Parallel execution and performance parameters, Speedup, Efficiency and Scalability, Amdahl's Law and Gustafson's Law, Multiprocessors and Multi computers, Shared memory vs Distributed memory systems, Flynn's Taxonomy (SISD, SIMD, MISD, MIMD).

Unit 2	Parallel Algorithms	10 hours

PRAM Algorithms, Parallel Reduction, Prefix Sum, List Ranking, Preorder Tree Traversal, Merging Two Sorted Lists, Graph Coloring. Performance Analysis.

Unit 3	Message Passing and Distributed Memory Programming (MPI)	10 hours

Distributed System, Message Passing model and distributed memory concept, MPI architecture and execution model, MPI point-to-point communication, Blocking and Non-blocking communication.
Hands-on/Case Study: Implementation of MPI programs.

Unit 4	Case Study of GPU Systems	10 hours

DGX architecture (A100 / H100 GPUs), NV Link and high-speed interconnects, CUDA software ecosystem, AI and deep learning workloads.

Total Lecture Hours	45 hours

Text Books

1. Michael J. Quinn, Parallel Programming in C with MPI and OpenMP, McGraw-Hill.
2. Peter Pacheco, An Introduction to Parallel Programming, Morgan Kaufmann.

Reference books:

1. Kai Hwang, Advanced Computer Architecture: Parallelism, Scalability, Programmability, McGraw-Hill.
2. John L. Hennessy and David A. Patterson, Computer Architecture: A Quantitative Approach, Morgan Kaufmann.
3. NVIDIA, CUDA C Programming Guide.
4. Grama, Gupta, Karypis, Kumar, Introduction to Parallel Computing, Pearson.

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
60	15	75	150
75			

Course Code: CS315L	Course Name: Cryptography and Network Security	L	T	P	C
		3	0	0	3

Pre-requisite: NIL

Course Objectives:

1. To introduce the fundamental concepts of information security, cryptography, and network security models.
2. To develop an understanding of symmetric and asymmetric cryptographic algorithms and key management techniques.
3. To study data integrity, authentication mechanisms, and digital signature techniques used in secure communication.
4. To understand secure communication protocols and web security mechanisms used in modern networks.
5. To analyze network security architectures such as IPsec, VPNs, and firewall systems for protecting network infrastructure.

Course Outcome: After completion of the course, the student will be able to

1. Explain the fundamental concepts of cryptography, security goals, types of attacks, and classical cryptographic techniques used in information security.
2. Analyze symmetric and public-key cryptographic algorithms, including AES, DES, RSA, and Diffie–Hellman, along with their applications in secure communication.
3. Apply hash functions, message authentication codes, digital signatures, and authentication mechanisms to ensure data integrity and secure communication.
4. Evaluate secure communication protocols and web security technologies such as SSL/TLS, HTTPS, SSH, and firewall systems.
5. Analyze the IPsec architecture, security associations, authentication headers, and VPN mechanisms used to secure IP-based communication networks.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	1	1	1	1	1	-	1	-	1
CO2	3	2	2	2	2	0	1	-	1	-	1
CO3	2	3	2	2	2	1	2	-	1	-	1
CO4	2	2	2	2	3	1	2	-	1	-	1
CO5	2	3	2	2	3	1	1	-	1	1	1

Unit 1 | Foundations of Cryptography & Security | 9 hours

Introduction to information and network security, Need for security; real-world incidents and impacts, Security goals: confidentiality, integrity, availability, authenticity, accountability, Types of security attacks Security services and mechanisms, Model for network security, Cryptography introduction, Symmetric vs asymmetric cryptography (conceptual overview), Classical substitution and transposition ciphers; limitations, Steganography: concept and simple modern examples

Unit 2 | Symmetric and Public-Key Cryptography | 9 hours

Symmetric key ciphers: design goals, confusion and diffusion, Block cipher principles: Feistel and substitution–permutation structures, DES: structure, key size, weaknesses and deprecation, Triple-DES (brief): role as a legacy enhancement. AES: key sizes, high-level round structure, industry applications, Overview of other symmetric ciphers: Blowfish, RC5, IDEA Block cipher modes, Stream ciphers, RC4 and reasons for its deprecation Public-key cryptography: problem of key distribution, trapdoor one-way functions,RSA algorithm: conceptual key generation, encryption, decryption, signatures, Diffie–Hellman key exchange, Knapsack-style public-key schemes. Suggested Practical Exercise: Applied Cryptography with AES, RSA, and Diffie–Hellman Using Python/OpenSSL

Unit 3 | Data Integrity & Authentication | 9 hours

E-mail security requirements and threats (spoofing, tampering, phishing), Pretty Good Privacy (PGP): keys, web of trust, encrypting and signing Cryptographic hash functions: properties and applications, Secure Hash Algorithm : SHA-2/SHA-512; Why MD5 and SHA-1 are considered broken/insecure. Message Authentication Codes (MACs): purpose and security requirements, HMAC, CMAC Digital signatures: goals (integrity, authentication, non-repudiation), Conceptual view of RSA/ElGamal-style signature schemes Authentication procedures: challenge–response, signed tokens, certificate-based login

Case study on password security, key-stretching methods, trusted software updates, and digital code-signing as used in real-world security systems.

Unit 4	**Secure Protocols & Web Security**	**9 hours**

Email security, S/MIME: certificate-based email security; integration with PKI, Web security considerations: sessions, cookies, basic browser threat landscape Overview of SSL and TLS evolution, TLS handshake: certificate verification, cipher suites, key establishment (conceptual) HTTPS: HTTP over TLS, Secure Shell (SSH): secure remote login, tunneling, key-based authentication, Firewalls: packet-filtering vs stateful, perimeter vs host firewalls, basic rule examples

Unit 5	**IP Security**	**9 hours**

IP security motivation: limitations of link-layer and application-layer only, IPsec overview and architecture: policies, Security Associations (SAs), Authentication Header (AH): what it protects, use cases, NAT issues (high level), Encapsulating Security Payload (ESP): confidentiality, integrity, anti-replay, Transport mode vs tunnel mode; host-to-host vs site-to-site VPNs, IPsec key management: high-level Internet Key Exchange (IKE) concepts

	Total Lecture hours	**45 hours**

Textbook:

1. William Stallings, "Cryptography and Network Security: Principals and Practice", Pearson Education.
2. Behrouz A. Frouzan: Cryptography and Network Security, McGraw Hill .
3. C K Shyamala, N Harini, Dr. T.R.Padmnabhan Cryptography and Security ,Wiley

Reference Books:

1. Bruce Schiener, "Applied Cryptography". John Wiley & Sons
2. Bernard Menezes," Network Security and Cryptography", Cengage Learning.
3. Atul Kahate, "Cryptography and Network Security", McGraw Hill

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
60	**15**	**75**	**150**
75			

Course Code: HS205L	**Course Name: Aptitude - 4**	**L**	**T**	**P**	**C**
		1	**0**	**0**	**1**

Pre-requisite: NIL

Course Objectives:

1. To gain a strong foundation in basic mathematical concepts, and become proficient in performing calculations, solving equations, and manipulating numerical data.
2. To apply quantitative methods effectively in interpreting data, optimizing solutions, and making informed decisions in academic, professional, and real-world engineering contexts.
3. To build a strong foundation in basic mathematical concepts including arithmetic, algebra, geometry, mensuration, trigonometry, and data interpretation required for engineering and analytical problem-solving.
4. To prepare for various competitive exams, such as aptitude tests for job placements, college admissions tests for higher education (GATE/CAT/GRE/GMAT) and government entrance exams. The course will provide the necessary knowledge and practice to perform well in these exams.

Course Outcome: After completion of the course, the student will be able to

1. Gain a strong foundation in mathematical concepts and demonstrate proficiency in calculations, equation solving, and numerical data manipulation, with the ability to apply these skills to solve mathematical and engineering problems.
2. Apply the principles of permutation, combination, and probability to analyze uncertainty and solve counting and chance-based problems relevant to mathematical and engineering applications.
3. Interpret and analyse data presented in various formats such as tables, graphs, and charts to support statistical analysis and informed decision-making.
4. Apply geometric principles and trigonometric relationships to analyze shapes, angles, and measurements; solve problems involving distances, heights, areas, and volumes; interpret diagrams and spatial data; and use these concepts effectively in mathematical, scientific, and engineering applications.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	1		1		1					2
CO2	3	1		1		1					2
CO3	3	1		1		1					2
CO4	3	1		1		1					2

Unit1	Number Systems-II and Algebra	3 hours

Types of numbers: (Surds indices, Logarithms, Progressions, Factors, Factorizations, Number of Zeros, Algebraic Identities, Linear and Quadratic Equations.

Unit 2	Permutation, Combination and Probability	3 hours

Set theory, Definition and Introduction of permutation and combination, Fundamental principle of counting by "AND" or "OR" rule, Forming of Numbers, Words, and Team; Problems related to linear and nonlinear arrangement. Application based on Geometry, Match, handshake, Chess board, Whole number, and Natural number solution, etc. Distribution: - Identical & Distinct, Basic concepts, definition and terminology related to probability, Problem based on Dice, Coins, Playing cards, Marbles, etc. Conditional Probability – Bayes theorem.

Unit 3	Data Interpretation, Data Sufficiency and Statistics	3 hours

Data Interpretation: Tables (Simple & Complex), Bar Graphs, Line Graphs, Pie Charts, Histograms, Radar graph, Caselets, Mixed Charts, Missing Data Problems, Comparison & Ratio Analysis, Percentage Change Analysis, Time-Based Data Sets, Data Sufficiency.

Statistics: Measures of Central Tendency (Mean, Median, Mode), Measures of Dispersion (Range, Mean Deviation, Variance, Standard Deviation), Averages & Weighted Mean, Quartiles, Deciles & Percentiles.

Unit 4	Geometry, Mensuration and Trigonometry	3 hours

Geometry: Basics of Geometry: Lines and Angles, Triangles (Types, Properties, Theorems), Congruence of Triangles, Similarity of Triangles, Quadrilaterals (Parallelogram, Rectangle, Square, Rhombus, Trapezium), Polygons, Circles (Chord, Tangent, Secant, Angles in a Circle). **Mensuration** – 2D Figures (Area & Perimeter), Mensuration – 3D Figures (Surface Area & Volume). **Trigonometry:** Introduction to Trigonometry, Trigonometric Ratios (sin, cos, tan, cosec, sec, cot), Trigonometric Identities, Values of Trigonometric Functions (Standard Angles), Trigonometric Equation, Heights and Distances.

Unit 5	Flowcharts and Gamified Aptitude	3 hours

Number Puzzles, Crypt Arithmetic, Flowcharts, Geo-Sudoku, Digit Challenge, Switch Challenge, Motion Challenge, Grid Challenge.

	Total Lecture hours	15 hours

Textbook

1. Booklet provided by the Aptitude Team, Training Division -CRPC, KIET

Reference Books

1. "Quantitative Aptitude for Competitive Examinations" by R.S. Aggarwal, S. Chand Publication.
2. "How to Prepare for Quantitative Aptitude for the CAT" by Arun Sharma, TMH Publication.
3. "Magical Book on Quicker Maths" by M. Tyra, Published by BSC Publishing Co. Pvt. Ltd
4. "Quantitative Aptitude Quantum CAT" by Sarvesh K. Verma, Arihant Publication.

Reference Websites:

1. https://www.indianbix.com
2. https://www.prepinsta.com
3. https://www.talentsprint.com
4. https://www.careeride.com
5. https://www.smartkeeda.com

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	25	25	50
25			

Open Elective-I Courses Detail Syllabus – VI Sem

Course Code: MA202O	Course Name: Reliability Theory and Maintenance	L	T	P	C
		2	0	0	2

Pre-requisite: 10+2

Course Objectives:

1. The objective of this course is to familiarize the graduate engineers with the concept of Reliability.
2. To equip students with techniques for reliability modeling, maintenance planning, and optimization of system availability and maintainability.

Course Outcome:

1. Understand the concept of Reliability of system with failure and distribution function.
2. Apply the concept of reliability on redundant system.
3. Employ the concept of reliability on complex system.
4. Understand the knowledge of Modeling, Maintainability and Availability of the system

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	2				1				2
CO2	2	2	2				1				2
CO3	2	2	2				1				2
CO4	2	2	2				1				2

Unit 1	Introduction to Reliability	08 hours

Definition of Reliability, Concept of Failure (Failure, failure rate, hazard rate), Types of failures: Early failure (infant mortality), Random failure, Wear-out failure; Failure density function, Bathtub Curve, Mean time to failure, Mean Time To Repair (MTTR), Mean Time Between Failures (MTBF), Estimation of Reliability function from failure data. Probability Distributions in Reliability: Binomial distribution, Poisson distribution, Exponential distribution, Weibull distribution

Unit 2	System Reliability	08 hours

Configuration of System components: Series Configuration, Parallel Configuration, Mixed configurations, k-out-of-n:G/F Configuration, Definition of Redundancy, Types of Redundancy: Active redundancy and Standby (passive) redundancy; Reliability evaluation of the system (Series systems, Parallel systems, Mixed system); Reliability of k-out-of-n:G/F system, reliability of Standby system (with perfect and imperfect switching).

Unit 3	Reliability of Complex Systems	07 hours

Definition of Complex system, Decomposition method, Conditional Probability approach, Minimal Path set method, Cut set method, Tie set method, Reliability Based Optimization, Introduction to Stochastic FEM

Unit 4	Modeling, Maintainability and Availability	07 hours

Reliability Block Diagrams (RBD), Fault tree analysis (FTA), Life Testing and Estimation: Parameter estimation (MLE method) and Goodness-of-fit tests, Markov models, Maintainability function, Availability function

	Total hours	30 hours

Textbook:

T.B-1: E. Balaguruswamy, "Reliability Engineering", Tata McGraw Hill.

T.B-2: D J Smith, Reliability, Maintainability & Risk, Butterworth-Heinemann

Reference Books:

1: Charles E Ebeling, "Reliability and Maintainability Engineering, Tata McGraw Hill

2: R.Billintan and R.N. Allan, "Reliability Evaluation of Engineering and Systems", Plenum Press.
3: K.C. Kapoor& L.R. Lamberson, "Reliability in Engineering and Design", John Wiely and Sons.
4: G.H.Sandler, "System Reliability Engineering", Prentice Hall.

Mode of Evaluation:

MSE	CA	ESE	Total
40	10	50	100

Continuous Assessment (CA): Video Assignments/ Assignment/ Quiz/ Project/ Seminar/ Group Discussion/ MOOC Course etc.

Course Code: EC237O	Course Name: Introduction to Robotic Systems	L	T	P	C
		2	0	0	2

Course Objectives:

1. To understand fundamentals of robotic systems, components and configurations.
2. To analyze robot kinematics, including forward/inverse kinematics and coordinate transformations.
3. To develop mathematical models of robot manipulators using DH parameters, Jacobian, and dynamic approaches.
4. To apply AI techniques (ANN, RL, Computer Vision) in robotic systems for perception, control, and decision-making.

Course Outcome: After completion of the course, the student will be able to

1. Explain and classify different types of robots, components, DOF and configurations.
2. Apply forward and inverse kinematics and perform coordinate transformations using homogeneous matrices.
3. Develop kinematic and dynamic models of robotic manipulators using DH parameters, Jacobian, and dynamic equations, and analyze system behavior.
4. Implement AI-based solutions such as: Object detection using OpenCV & YOLO and ANN-based control and trajectory generation.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	2	-	-	2	-	2	2	-	2
CO2	2	2	2	-	-	2	-	2	2	-	2
CO3	3	3	3	-	-	3	-	3	3	-	2
CO4	3	2	2	-	3	3	3	2	2	3	3

Unit 1	Introduction to Robotic Systems	7 hours

Fundamentals of Robotics: Types of Robots, Robot Components, Manipulator, Reference Frames, Degrees of Freedom, Joints of a Robot, Robot Configurations, End Effector.

Unit 2	Kinematic Transformations	7 hours

Kinematics Equations: Forward Kinematics & Inverse Kinematics, Workspace

Coordinate Frames and Homogeneous Transformation: Coordinate Transformations, Rotation, Translation, Composition of homogeneous transformations

Unit 3	Kinematic Model for Robot Manipulator	8 hours

Denavit-Hartenberg Procedure & Coordinates Frames, 3-DOF & 6-DOF Manipulator, 4-Axis SCARA Robot

Manipulator Jacobian

Dynamics of Manipulator: Lagrange-Euler approach, Newton-Euler approach, Multiple Degree of Freedom

Unit 4	AI-Integrated Robotic Systems	8 hours

Introduction to AI in Robotics: Role of AI in modern robotics, Difference between Traditional vs Intelligent robots

AI for Robot Perception: Sensor data processing, Basics of computer vision using OpenCV, Object detection

Total Lecture hours **30 hours**

Textbook

1. Subir Kumar Saha, "Introduction to Robotics", Tata McGraw-Hill Education, 2014.
2. Richard Paul, Robot Manipulators: Mathematics, Programming and Control, MIT Press, 1981
3. Robert Shilling, Fundamentals of Robotics, Prentice-Hall , 2003
4. Laxmidhar Behera and Indrani Kar, "Intelligent Systems and Control", Oxford University Press, Nov 2009.

Reference Books

1. M. Felix Orlando, Ashish Dutta, Anupam Saxena, Laxmidhar Behera, Tomoya Tamei and Tomohiro Shibata, "Manipulability Analysis of Human Thumb, Index and Middle Finger in Cooperative 3D Rotational Movements of a Small Object", Robotica, vol. 31, pp. 797-809, 2013.
2. M. Felix Orlando, Laxmidhar Behera, Tomayo Tamei, Tomohiro Shibata, Ashish Dutta and Anupam Saxena," On Redundancy Resolution of the Human Thumb, Index and Middle Fingers in Cooperative Object Translation," Robotica, vol. 35, pp. 1992-2017, 2016.

Mode of Evaluation:

MSE	CA	ESE	Total
40	10	50	100

CA1- Report Submission on Addverb Technologies Visit**, CA2-**Assignment Based**, CA3-**Attendance

Course Code: EE214O	Course Name: AIoT for Smart Energy	L	T	P	C
		2	0	0	2

Pre-requisite: Basic knowledge of Electronics, Programming in Python/C, and Mathematics.

Course Objectives:

1. To introduce the fundamentals of Artificial Intelligence of Things (AIoT), its architecture in the context of smart energy systems and to design IoT sensor networks for real-time energy monitoring, data acquisition, edge-level processing.
2. To develop the ability to apply machine learning and AI algorithms for energy demand forecasting, anomaly detection, renewable energy prediction and smart grid technologies, cloud platforms, and the future roadmap of AIoT in sustainable energy.

Course Outcome: After completion of the course, the student will be able to

1. Understand the fundamentals of AIoT architecture, communication protocols, and their role in the smart energy ecosystem.
2. Design and analyze IoT sensor networks for real-time energy monitoring and data acquisition.
3. Apply machine learning and AI algorithms to predict energy demand and detect anomalies in energy systems.
4. Evaluate smart grid technologies, cloud platforms, renewable energy integration, and AIoT-based optimization strategies.

Unit 1	Foundations of AIoT and Smart Energy	7 hours

Overview of AIoT: convergence of Artificial Intelligence and Internet of Things. AIoT system architecture: Perception, Network, Platform and Application layers. Introduction to smart energy systems, smart meters and Advanced Metering Infrastructure (AMI). Key energy metrics: kWh, Power Factor and Load Factor. Communication protocols: MQTT, Zigbee and LoRaWAN. Edge computing versus cloud computing in energy applications.

Unit 2 IoT Sensors, Devices and Connectivity	**8 hours**
Energy sensors: current transformers, voltage sensors, irradiance sensors and smart plugs. Microcontroller platforms: Arduino and ESP32 for energy monitoring. Wireless communication: Zigbee and Bluetooth for short range, LoRaWAN for long range, NB-IoT for smart grid applications. Introduction to Edge AI: TinyML and TensorFlow Lite for on-device processing. Basic data pre-processing: noise filtering, handling missing values and time-series data characteristics.	
Unit 3 Artificial Intelligence for Energy Management	**8 hours**
Overview of supervised and unsupervised machine learning with energy examples. Energy demand forecasting using Linear Regression, Random Forest and LSTM networks. Anomaly detection for electricity theft and equipment faults using Isolation Forest and Autoencoders. Solar and wind power prediction using weather data and ML models. Introduction to Reinforcement Learning for smart thermostat and battery scheduling optimization.	
Unit 4 Smart Grids, Cloud Platforms and Future Applications	**7 hours**
Smart grid architecture: evolution from traditional grid to smart grid, demand response and Virtual Power Plants. Cloud platforms: AWS IoT Core, Google Cloud IoT and open-source tools such as Node-RED and ThingSpeak. AIoT for building energy management and EV smart charging. Introduction to Vehicle-to-Grid (V2G) technology. Emerging trends: Federated Learning for energy data privacy, Blockchain for peer-to-peer energy trading and role of AIoT in achieving Net Zero targets.	
Total Lecture Hours	**30 hours**

Textbook:

1. A. Bahga and V. Madisetti, *Internet of Things: A Hands-On Approach,* VPT, 2014.
2. F. Chollet, *Deep Learning with Python,* Manning Publications, 2021.
3. J. Ekanayake, K. Liyanage, J. Wu, A. Yokoyama, N. Jenkins, *Smart Grid: Technology and Applications,* Wiley, 2012.
4. A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow,* 3rd ed., O'Reilly Media, 2022.

Reference Books:

5. S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach,* 4th ed., Pearson, 2020.
6. R. H. Shumway and D. S. Stoffer, *Time Series Analysis and Its Applications,* Springer, 2017.
7. M. Schwartz, *Internet of Things with ESP8266,* Packt Publishing, 2016.
8. P. J. Sadalage and M. Fowler, *NoSQL Distilled,* Addison-Wesley, 2012.

Mode of Evaluation:

MSE	CA	ESE	Total
40	10	50	100

Course Code: ME229O	**Course Name: Advanced Engineering Materials**	**L**	**T**	**P**	**C**
		2	**0**	**0**	**2**

Pre-requisite: Basic knowledge of Physics and Chemistry

Course Objectives:

1. To introduce students to the classification and properties of advanced engineering materials.
2. To develop understanding of modern materials used in aerospace, biomedical, electronics, and energy applications.
3. To familiarize students with Semiconductors,smart materials, nanomaterials, and composite materials.
4. To enable students to correlate material properties with engineering applications and sustainability requirements.

Course Outcome: After completion of the course, the student will be able to

1. Explain the structure, properties, and classification of advanced engineering materials.
2. Analyze the applications of composite, smart, nano, and biomaterials in modern engineering systems.
3. Select suitable advanced materials for specific industrial and technological applications.
4. Understand sustainable and emerging material technologies for future engineering challenges.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	-	-	-	-	-	2
CO2	2	3	2	1	1	-	-	-	-	-	2
CO3	2	3	3	1	2	-	-	-	-	-	2
CO4	2	2	1	-	-	1	1	-	-	-	3

Unit 1	Composite materials	7 hours

Introduction to composite materials and their Classification. Fiber reinforced composites (FRP, CFRP, GFRP)
Metal Matrix Composites (MMC), Ceramic Matrix Composites (CMC), Manufacturing methods and applications Properties and applications of composites in automotive and aerospace sectors.

Unit 2	Semiconductor and Smart Materials	8 hours

Introduction to conductors, semiconductors, and insulators, Crystal structure of semiconductors, Energy band theory, Intrinsic and extrinsic semiconductors, Carrier concentration, mobility, conductivity, Fermi level concept Smart materials: shape memory alloys, piezoelectric materials, magnetostrictive materials, and electroactive materials. Applications of smart materials in sensors and actuators.

Unit 3	Nanomaterials and Biomaterials	8 hours

Introduction to nanomaterials and nanotechnology. Synthesis and characterization of nanomaterials. Carbon nanotubes, graphene, nanoparticles, and Nano coatings. Applications of nanomaterials in electronics, energy storage, and environmental engineering. Biomaterials: metallic, ceramic, polymeric, and composite biomaterials. Applications in medical implants and tissue engineering.

Unit 4	Advanced Functional Materials and Sustainable Materials	7 hours

Semiconductors and electronic materials. Superconducting materials and optical materials. Materials for renewable energy systems such as solar cells, batteries, and fuel cells. Eco-friendly and sustainable materials. Recycling and green processing of materials. Recent trends and future scope of advanced engineering materials.

Total Lecture Hours	30 hours

Textbook:

1. Callister, W.D. and Rethwisch, D.G., Materials Science and Engineering: An Introduction, Wiley India.
2. Raghavan, V., Materials Science and Engineering, Prentice Hall India.
3. Donald R. Askeland and Wendelin J. Wright, The Science and Engineering of Materials, Cengage Learning.
4. William F. Smith and Javad Hashemi, Foundations of Materials Science and Engineering, McGraw Hill.

Reference Books:

1. Charles P. Poole Jr. and Frank J. Owens, Introduction to Nanotechnology, Wiley.
2. M.F. Ashby, H. Shercliff, and D. Cebon, Materials: Engineering, Science, Processing and Design.
3. S.O. Kasap, Principles of Electronic Materials and Devices, McGraw Hill.
4. K.K. Chawla, Composite Materials: Science and Engineering, Springer

Mode of Evaluation:

MSE	CA	ESE	Total
40	10	50	100

Course Code: MS203O	Theory Course Name: Business Strategy	L	T	P	C
		2	0	0	2

Pre-requisite: NIL

Course Objectives

1. To develop strategic thinking from a business perspective
2. To analyse industry dynamics and strategic positioning.
3. To enable students to design business strategies in technology-driven environments.

Course Outcome (COs)

CO No.	Statement of Course Outcome
After the completion of the course, the student will be able to:	
CO 1	**Apply** strategic tools to business scenarios
CO 2	**Analyze** competitive environments and firm positioning
CO 3	**Evaluate** strategic choices and business models
CO 4	**Assess** the impact of innovation and emerging technologies on business strategy.

CO-PO Mapping (scale 1: low, 2: Medium, 3: High)

Course Outcomes (COs)	Programme Outcomes (POs)					Additional Programme Outcomes (APOs)	
	PO1	PO2	PO3	PO4	PO5	APO 1	APO 2
CO 1	1	2	2	2	2	3	3
CO 2	3	3	3	3	3	3	3
CO 3	2	3	3	3	2	3	3
CO 4	3	3	3	3	3	3	3

DETAILED SYLLABUS

Unit 1	Foundations of Business Strategy	Hours 7

Strategy vs Business Model, Levels of strategy, Strategic management process, Concept of Competitive Advantage, Stakeholders in Business Strategy

Unit 2	Industry and Competitive Analysis	Hours 8

Industry Structure and Dynamics, Porter's Five Forces Model, Competitor Analysis , Market Segmentation and Targeting (basic concepts), Strategic Positioning, Tools: SWOT Analysis, PESTLE

Unit 3	Strategic Choices and Business Models	Hours 8

Generic Strategies: Cost Leadership vs Differentiation, Focus Strategy (Niche markets), Growth Strategies: Expansion, Diversification, Innovation, Role of Technology and AI in shaping business strategy.

Unit4	Strategy Implementation & Contemporary Issues	Hours 7

Strategy Implementation: Organizational Structure and Culture, Leadership and Change Management, Resource Allocation. Contemporary Strategic Issues: Digital Transformation, Artificial Intelligence in Strategy, Sustainability and Corporate Social Responsibility (CSR), Ethics in Strategic Decision-Making.

Total Lecture hours	30 hours

Textbook

1. David & David – *Strategic Management: Concepts and Cases*
2. Michael Porter – *Competitive Strategy*
3. Osterwalder & Pigneur – *Business Model Generation*
4. Selected case studies (Harvard/industry-based simplified versions)

Reference Books

1. https://www.ebookbou.edu.bd/Books/Text/SOB/MBA/mba_3311/Unit-07.pdf
2. https://nptel.ac.in/courses/110/108/110108047/
3. https://swayam.gov.in/nd2_imb19_mg08/preview

Mode of Evaluation

Continuous Assessment (CA): Case Presentation/ Activities/ Assignment / Quiz / Project/ MOOC etc.

MSE	CA	ESE	Total
40	10	50	100

Professional Elective-IV Courses Detail Syllabus – VI Sem

Course Code:IT316E	Course Name: Agentic AI Foundation and Applications	L	T	P	C
		2	0	4	4

Pre-requisite: Artificial Intelligence Fundamentals, Python Programming, Basic Machine Learning & NLP

Course Objectives:

This course provides learners with a comprehensive theoretical understanding of **Agentic Artificial Intelligence**, focusing on intelligent agent concepts, architectures, reasoning and planning mechanisms, learning strategies, and decision-making processes. The course also emphasizes real-world applications such as task automation, decision support, conversational agents, and workflow orchestration using modern AI frameworks and APIs.

Course Outcome: By the end of this course, learners will be able to

1. Understanding the core principles of **Agentic AI**, including intelligent agents, environments, and decision-making models.
2. Analyzing agent architectures, reasoning, planning, and learning mechanisms used in autonomous systems.
3. Evaluating single-agent and multi-agent interactions, coordination strategies, and communication models.
4. Applying agent-based concepts to real-world application domains such as automation, recommendation, and decision support.
5. Examining ethical, safety, governance, and societal implications of deploying autonomous and agentic AI systems.

CO-PO Mapping (Scale1:Low, 2:Medium, 3:High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	2	-	2	-	-	-	2
CO2	3	3	2	2	3	-	-	-	-	-	2
CO3	2	3	3	2	3	-	-	2	-	-	2
CO4	2	2	3	2	3	-	-	2	2	-	2
CO5	2	2	2	2	2	3	3	2	2	-	3

Unit 1	Foundations of Agentic Artificial Intelligence	18 hours

Introduction to Artificial Intelligence and evolution towards Agentic AI; concept of intelligent agents; agent–environment interaction; types of agents (simple reflex, model-based, goal-based, utility-based, learning agents); rationality and performance measures; characteristics of agentic systems; comparison of traditional AI vs agent-based AI

Case Study: Design and analysis of an intelligent personal assistant as a goal-based agent.

Unit 2	Agent Architectures and Reasoning Mechanisms	18 hours

Agent architectures (reactive, deliberative, hybrid); knowledge representation for agents; reasoning techniques (rule-based reasoning, logical reasoning, probabilistic reasoning); planning concepts; goal formulation and action selection; handling uncertainty in decision-making.

Case Study: Agent-based decision-making system for autonomous traffic signal control.

Unit 3	Learning and Adaptation in Intelligent Agents	18 hours

Learning agents and feedback mechanisms; reinforcement learning fundamentals for agents; exploration vs exploitation; policy learning; adaptation and self-improving agents; memory, experience replay, and environment modeling; limitations of learning-based agents.

Case Study: Learning-based recommendation agent for personalized content delivery.

Unit 4	Development of Agentic AI Applications	21 hours

Multi-agent system (MAS) concepts; agent communication languages; LLM-based task agents; coordination, cooperation, and negotiation strategies; distributed problem solving; agent societies; real-world applications of agentic AI in automation, robotics, finance, healthcare, and smart systems.

Case Study: Multi-agent coordination in smart grid energy management.

Unit 5	Ethics, Safety, and Governance of Agentic AI	15 hours

Ethical challenges in autonomous systems; fairness, transparency, and explainability; safety and reliability of agentic systems; human-in-the-loop models; governance frameworks for AI agents; accountability and regulatory considerations; future directions of Agentic AI.

Case Study: Ethical analysis of autonomous decision-making agents in healthcare systems.

	Total Lecture Hours	90 hours

Textbook:

1. Artificial Intelligence: A Modern Approach – Stuart J. Russell & Peter Norvig
2. Multi Agent Systems: Technologies and Applications towards Human-Centered – Shibakali Gupta, Indradip Banerjee & Siddhartha Bhattacharyya

Reference Books:

3. Agentic AI: Theories and Practices – Edited by Ken Huang
4. AI Agents in Action (or Second Edition) – Micheal Lanham

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: IT312E	Course Name: Cloud Security Principles & Practices	L	T	P	C
		2	0	4	4

Pre-requisite: AWS Solution Architect

Course Objectives:

The primary objective of this course is to equip learners with advanced cloud architecture, large-scale system design, governance, and migration skills required to architect complex, enterprise-grade solutions on AWS, aligned with the AWS Certified Solutions Architect – Professional certification.

Course Outcome: By the end of this course, learners will be able to:

1. Design enterprise-scale, multi-account AWS architectures with governance and compliance.
2. Architect complex hybrid and multi-region systems with high availability and fault tolerance.
3. Plan and execute large-scale cloud migration and modernization strategies.
4. Implement advanced DevOps, security, and cost governance practices at organizational scale.
5. Analyze ambiguous real-world scenarios and justify architectural decisions using AWS best practices.

CO-PO Mapping (Scale1:Low,2:Medium,3:High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	3	-	-	2	2	3	3
CO2	3	3	2	2	3	-	-	2	2	2	3
CO3	3	3	2	3	3	-	-	2	2	3	2
CO4	3	2	3	2	3	2	3	3	2	2	2
CO5	3	3	3	-	3	-	-	2	3	3	3

Unit1	Enterprise Cloud Architecture & Governance	18 hours

Enterprise cloud adoption frameworks.
AWS Organizations and multi-account strategy: landing zones, account vending, SCPs.
Control Tower architecture and governance automation.
Centralized logging, monitoring, and audit architectures.
Enterprise identity federation and cross-account access models.
Case Study: Designing a Secure Multi-Account AWS Environment for a Large Enterprise

Unit 2	Advanced Networking & Hybrid Architectures	18 hours

Complex VPC architectures: hub-and-spoke, transit gateway, shared services VPC.
Hybrid connectivity patterns: site-to-site VPN, client VPN, Direct Connect architecture.
Multi-region networking, latency-based routing, and failover strategies.
Private connectivity to AWS services using VPC endpoints and PrivateLink.
Network security at scale: centralized inspection and traffic control.
Case Study: Hybrid On-Premises to AWS Architecture using Transit Gateway and Direct Connect

Unit 3	Large-Scale Migration & Modernization Strategies	18 hours

Cloud migration frameworks and readiness assessment.
Migration strategies: rehost, replatform, refactor, retain, retire.
Data migration at scale: DMS, Snowball, Snowmobile (architecture perspective).
Application modernization: monolith to microservices, containers, serverless.
Managing technical debt and phased migration planning.
Case Study: Enterprise Application Migration from On-Premises to AWS with Minimal Downtime

Unit 4	Advanced DevOps, Security & Reliability Engineering	18 hours

Enterprise CI/CD architecture across multiple teams and accounts.
Advanced deployment strategies: multi-region blue-green, canary at scale.
Reliability engineering: chaos engineering, fault injection, resilience testing.
Security at scale: centralized IAM, KMS key strategies, secrets governance.
Compliance automation: continuous compliance monitoring and remediation.
Case Study: Highly Resilient Multi-Region SaaS Platform with Automated Recovery

Unit 5	Cost Governance, Optimization & Professional Exam Capstone	18 hours

Cost governance models: chargeback, showback, and budget enforcement.
Advanced cost optimization: Savings Plans, Reserved Instances, Spot Fleets.

Performance optimization at scale: caching, asynchronous processing, event-driven design.
Architectural trade-off analysis: cost vs performance vs reliability vs security.
Professional-level exam scenarios and whiteboard architecture justification.

Case Study: Designing a Global, Multi-Region, Enterprise-Grade AWS Architecture (Professional-Level)
Digital Course: AWS CloudOps Engineer - Introduction to Cloud Operations
Exam Readiness: AWS Certified Solutions Architect – Professional (SAP-C02)
Exam Prep Official Practice Question Set: AWS Certified Solutions Architect – Professional (SAP-C02)
Exam Prep Enhanced Course: Solutions Architect – Professional (SAP-C02) Learning Plan (includes labs)

Total Lecture	**90 hours**

Textbook:

1. Kavis, M. Architecting the Cloud: Design Decisions for Cloud Computing Service Models.
2. Wittig, A. & Wittig, M. Amazon Web Services in Action (Advanced reference).

Reference Books:

1.Beyer et al. Site Reliability Engineering
2.Kim et al. The DevOps Handbook
3.AWS Architecture Center – Official Whitepapers
4.AWS Well-Architected Framework (Advanced Lens)

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
80	**20**	**100**	**200**
100			

Course Code: IT320E	**Course Name: Advanced Cloud Architecture**	**L**	**T**	**P**	**C**
		2	**0**	**4**	**4**

Pre-requisite: Cloud Architecture Foundation / AWS Solutions Architect – Associate level knowledge

Course Objectives:

- To design enterprise-grade cloud architectures with high availability and fault tolerance.
- To implement multi-region, disaster recovery, and hybrid cloud strategies.
- To design secure, compliant, and cost-optimized cloud environments.
- To architect microservices and container-based distributed systems.
- To evaluate architectural trade-offs using cloud well-architected principles.

Course Outcome: By the end of this course, learners will be able to

CO1: Design multi-tier, highly available, and fault-tolerant cloud architectures.
CO2: Architect hybrid and multi-cloud environments with secure connectivity.
CO3: Implement containerized and microservices-based architectures.
CO4: Design disaster recovery and business continuity strategies.
CO5: Analyze performance, security, and cost trade-offs in enterprise cloud systems.

CO-PO Mapping (Scale1:Low,2:Medium,3:High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	2	3	-	-	2	2	2	3
CO2	3	3	3	2	3	-	-	2	2	2	3
CO3	3	3	3	2	3	-	-	2	2	2	3
CO4	3	3	3	2	3	-	-	2	3	2	3
CO5	3	3	3	2	3	-	-	2	2	2	3

Unit1	Enterprise Cloud Architecture & Design Patterns	18 hours
Advanced cloud design principles, AWS Well-Architected Framework, architectural trade-offs, distributed system fundamentals, CAP theorem, scalability patterns, stateless vs stateful systems, caching strategies, content delivery networks, API design principles. Case Study: Designing a globally distributed e-commerce platform.		
Unit 2	**High Availability, Disaster Recovery & Multi-Region Architecture**	**18 hours**
Multi-AZ and multi-region deployment strategies, RTO & RPO planning, backup and restore mechanisms, active-active vs active-passive architecture, database replication strategies, cross-region failover, global load balancing. Case Study: Multi-region deployment with automated failover.		
Unit 3	**Microservices, Containers & Kubernetes Architecture**	**18 hours**
Microservices architecture patterns, containerization concepts, Docker fundamentals, Kubernetes architecture, orchestration, service mesh basics, CI/CD integration, infrastructure as code (IaC), DevOps principles in cloud architecture. Case Study: Deploying scalable microservices using Kubernetes.		
Unit 4	**Hybrid, Multi-Cloud & Secure Architecture**	**18 hours**
Hybrid cloud design, VPN and Direct Connect concepts, identity federation, Zero Trust architecture, encryption strategies, secrets management, compliance frameworks, governance models, policy-driven architecture. Case Study: Secure hybrid cloud integration for enterprise ERP.		
Unit 5	**Performance Optimization, Cost Engineering & Observability**	**18 hours**
Performance tuning strategies, autoscaling policies, cost optimization frameworks, cloud pricing models, monitoring & observability tools, logging, tracing, incident response planning, reliability engineering principles (SRE basics). Case Study: Cost and performance optimization for production workload. AWS Cloud Quest: Serverless Developer Exam Readiness: Exam Guide for AWS Certified Developer - Associate (DVA-C02) Exam Prep Official Practice Question Set: AWS Certified Developer - Associate (DVA-C02) Exam Prep Enhanced Course: AWS Certified Developer - Associate (DVA-C02)		
	Total Lecture Hours	**90**

Textbook:

1. Wittig, M., & Wittig, A. (2021). Amazon Web Services in Action (2nd ed.). Manning.
2. Sbarski, P., Cui, Y., & Nair, A. (2022). Serverless Architectures on AWS (2nd ed.). O'Reilly.
3. Newman, S. (2021). Building Microservices (2nd ed.). O'Reilly.

Reference Books:

1. AWS Well-Architected Framework Documentation.
2. Google Cloud Architecture Frameword.
3. Microsoft Azure Architecture Center

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
80	**20**	**100**	**200**
100			

Course Code: CS323E	Course Name: Advanced iOS Frameworks & App Store Deployment	L	T	P	C
		2	0	4	4

Pre-requisite: iOS App Development & Swift UI

Course Objectives:

1. To develop Advanced User Interface Design and Native Hardware Frameworks
2. To build and Implement Robust Local Data Persistence and System Notifications

Course Outcome: After completion of the course, the student will be able to

CO1. To apply interactive SwiftUI interfaces using animations, gestures, and custom view modifiers.
CO2. To Develop iOS applications using Core Data and Realm for local data storage and management.
CO3. To apply audio and video playback features into SwiftUI applications.
CO4. To develop location-based services and interactive maps using Core Location and MapKit.
CO5. To Develop and debug a capstone iOS application by integrating UI enhancements, local databases, maps, media, and notifications.

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	2	3		3	-	-	-	-	-	2
CO2	3	3	3	2	3	-	-	-	-	-	2
CO3	2	2	3		3	-	-	-	-	-	2
CO4	3	3	3	2	3	-	-	-	-	-	2
CO5	3	3	3	2	3	-	-	-	-	-	2

Unit 1	Swift UI UI/UX Enhancements	18 hours

Basic Animations: Implementing simple implicit and explicit view animations.
Transitions & Gestures: Adding basic view transitions and handling common interactions like Tap, Drag, and Long Press.
Custom View Modifiers: Creating simple, reusable modifiers to apply consistent styles across views.

Unit 2	Local Databases - Core Data & Realm	18 hours

Core Data Essentials: Setting up simple entities and performing basic CRUD (Create, Read, Update, Delete) operations.
Introduction to Realm: Setting up a Realm database as a lightweight alternative to Core Data.
Data Saving: Saving and fetching local user data within SwiftUI views.

Unit 3	Basic Media & Audio-Video Playback	18 hours

Media Playback: Using basic framework tools to play audio and video files inside an app.
Playback UI: Building straightforward play/pause controls in SwiftUI.
Media Management: Importing and displaying basic audio/video assets cleanly.

Unit 4	Location Services & Maps	18 hours

Core Location Basics: Requesting location permissions and getting the user's current GPS coordinates.
MapKit Integration: Displaying a basic interactive map inside a SwiftUI view.
Map Annotations: Adding simple pins or markers to show specific locations on the map.

Unit 5	Notifications & Capstone Project	18 hours

Local Notifications: Triggering simple scheduled alerts and reminders on the user's device.
Project Planning: Combining UI enhancements, local databases, maps, and notifications into a single app idea.
Final Implementation: Building and debugging a simplified capstone application on the simulator.

	Total Lecture Hours	90 hours

Textbook:

1. iOS App Development Essentials (SwiftUI Edition) by Neil Smyth
2. SwiftUI Cookbook by Giordano Scalzo and Edgar N. Miranda

Reference Books:

1. Pro iPhone Development with SwiftUI: Designing Professional Apps with Apple's Frameworks by Wallace Wang (Apress)
2. AVFoundation Framework Reference by Apple Documentation & Developer Publications

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Course Code: IT321B	Course Name: Software Testing Techniques and Tools	L	T	P	C
		3	0	2	4

Pre-requisite: Programming Fundamentals, Data Structures, Software Engineering Concepts, Operating Systems (basic understanding), Problem Solving & Logical Reasoning

Course Objectives:

This course enables students to understand the fundamental principles, processes, and terminology of software testing. It helps them develop the ability to design effective test cases using functional and structural testing techniques. Students are introduced to advanced testing strategies and quality evaluation metrics to strengthen their analytical skills. The course provides practical exposure to modern automated testing tools and frameworks. It also familiarizes students with non-functional testing and contemporary industry practices relevant to real-world software development.

Course Outcome: After completion of the course, the student will be able to

1. Understand fundamental principles, terminology, and lifecycle of software testing.
2. Apply structural and functional testing techniques to design effective test cases.
3. Analyse software behaviour using advanced testing models and strategies.
4. Use modern automated testing tools for functional, UI, API, and regression testing.
5. Evaluate software quality using non-functional testing approaches and contemporary practices.

CO-PO Mapping(Scale1:Low, 2:Medium, 3:High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-
CO3	3	3	2	3	-	-	-	-	-	-	-
CO4	2	2	3	2	3	-	-	-	-	-	-
CO5	2	3	2	2	-	-	-	-	-	-	3

Unit1	Fundamentals of Software Testing	15 hours

Introduction to Software Testing: Purpose, Objectives, and Principles, Software Development Life Cycle (SDLC) & Testing Life Cycle (STLC), Verification & Validation, Error, Fault, Failure, Defect, Bug, Test Process & Test Planning, Test Case Design Concepts, Static Testing: Reviews, Walkthroughs, Inspections, Test Documentation & Test Artifacts, Test Oracles & Testability, Introduction to Software Quality Assurance (SQA).

Case Study 1: Test Case Design and Documentation
Objective: To design structured test cases based on given requirements.
Procedure:
• Study the requirement specification.
• Identify test scenarios.
• Design positive and negative test cases.
• Document inputs, expected outputs, and results.

Case Study 2: Static Testing – Review & Inspection
Objective: To perform static testing using review techniques to identify defects.
Procedure:
• Analyze given code/document.

• Identify defects without execution.
• Classify defects.

Unit 2	Functional & Structural Testing Techniques	15 hours

Functional Testing Techniques: Black Box Testing, Boundary Value Analysis (BVA), Equivalence Class Partitioning (ECP), Decision Table Testing, State Transition Testing, Use Case Testing, Structural Testing Techniques: White Box Testing, Control Flow Testing, Flow Graphs & Paths, Cyclomatic Complexity, Basis Path Testing, Data Flow Testing, Path Sensitization & Coverage Criteria.

Case Study 3: Black Box Testing Techniques
Objective: To apply Equivalence Class Partitioning (ECP) and Boundary Value Analysis (BVA) techniques for test case design.
Procedure:
• Identify input parameters.
• Partition inputs into equivalence classes.
• Identify boundary values.
• Execute test cases.

Case Study 4: Structural Testing & Coverage Analysis
Objective: To perform white box testing using control flow analysis and path coverage techniques.
Procedure:
• Draw Control Flow Graph.
• Compute cyclomatic complexity.
• Identify independent paths.
• Design path test cases.

Unit 3	Advanced Testing Strategies	15 hours

Regression Testing & Test Case Prioritization, Integration Testing Strategies, System Testing, Exploratory Testing, Risk-Based Testing, Mutation Testing, Test Coverage Metrics, Defect Management & Debugging, Test Adequacy & Test Effectiveness, Test Automation Concepts & Strategy, Continuous Testing & DevOps Testing Overview.

Case Study 5: Regression Testing
Objective: To validate software functionality after modifications using regression testing techniques.
Procedure:
• Identify impacted modules.
• Re-run test suites.

Case Study 6: Mutation Testing
Objective: To evaluate the effectiveness of test cases using mutation testing.
Procedure:
• Introduce controlled faults.
• Execute tests.

Unit 4	Test Automation & Modern Testing Tools	15 hours

Test Automation Fundamentals: Need for Automation, Automation Frameworks (Data-driven, Keyword-driven, Hybrid), Test Scripts & Test Suites, Modern Testing Tools: UI Testing Tools (e.g., Selenium Concepts), API Testing Tools (e.g., Postman Concepts), Test Management Tools, Continuous Integration Tools (Testing Perspective), Automated Regression Testing, Test Reporting & Logging.

Case Study 7: Selenium UI Automation Testing
Objective: To automate user interface testing using Selenium WebDriver.
Procedure:
• Configure Selenium WebDriver.
• Identify UI elements.
• Write automation scripts.

Case Study 8: Automated Regression Testing
Objective: To automate regression test scenarios using test suites.
Procedure:
• Develop test suites.
• Execute repeatedly.

Unit5	Non-Functional & Contemporary Testing	15 hours

Performance Testing Concepts, Load & Stress Testing, Security Testing Fundamentals, Usability Testing, Compatibility Testing, Reliability Testing, Database Testing Basics, Web & Mobile Application Testing Overview, Cloud Testing & Distributed Systems Testing, Contemporary Trends: Agile Testing, DevOps & Continuous Testing, AI-Assisted Testing (Conceptual Overview).

Case Study 9: Performance Testing Using JMeter
Objective: To evaluate system performance using load testing techniques.
Procedure:
• Create Test Plan.
• Configure Thread Group.
• Execute tests.

Case Study 10: Security & Usability Testing
Objective: To perform basic non-functional testing focusing on security and usability aspects.
Procedure:
• Identify vulnerabilities/usability issues.

Total Lecture Hours	75 hours

Textbook:

1. Boris Beizer, Software Testing Techniques, Dreamtech Press
2. Yogesh Singh, "Software Testing", Cambridge University Press, New York, 2012
3. Ron Patton, Software Testing, Sams Publishing
4. Paul Ammann & Jeff Offutt, Introduction to Software Testing, Cambridge University Press
5. Naresh Chauhan, Software Testing: Concepts and Tools, Oxford University Press (Indian Edition).

References:

1. Glenford J. Myers, The Art of Software Testing, Wiley
2. Rex Black, Foundations of Software Testing, Cengage Learning
3. Elfriede Dustin, Effective Software Testing, Addison-Wesley
4. Lisa Crispin & Janet Gregory, Agile Testing, Addison-Wesley
5. Aditya P. Mathur, Foundations of Software Testing, Pearson

Mode of Evaluation

Evaluation Scheme			
MSE	CA	ESE	Total Marks
80	20	100	200
100			

Practical Courses Detail Syllabus – VI SEM

Course Code: HS206P	Course Name: Soft Skills for Employability - 2	L	T	P	C
		0	0	2	1

Pre-requisite: Successful completion of Soft Skills Essentials I & II and Employability Skills – I (Verbal Ability).

Course Objectives:
To strengthen students' placement preparedness and develop employability skills by enhancing creativity, critical thinking, resume development, resume preparation, group discussion skills, and interview techniques.

Course Outcome: After completion of the lab, the student will be able to
CO1: Demonstrate interpersonal effectiveness through handling group discussions and professional presence through visumes.
CO2: Justify their resumes through effective professional presentation skills.
CO3: Apply interview techniques and feedback to successfully handle employment interviews.
CO4: Develop a professional digital identity by utilizing AI-enabled career tools, networking platforms, psychometric insights, and employability strategies to enhance career readiness.

CO-PO Mapping (scale 1: low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	1	-	2	3	-	2
CO2	-	-	-	-	-	-	-	-	3	-	2
CO3	-	-	-	-	-	-	-	1	3	-	2
CO4	-	-	-	-	-	1	-	-	2	-	2

List of Activities:

1. LLM Enabled Professional Self-Introductions using Mind-Mapping
2. Employability & Placement Process Overview
3. Resume Review Lab – Peer Review & Faculty Feedback
4. Group Discussion Practice Sessions
5. Personal Branding for Career Development – LinkedIn, AlmaConnect, E-Portfolio
6. Psychometric Test (MBTI / Locus of Control)
7. Preparation for Job Interviews – 4 Ts (Types, Tips, Tricks & Techniques)
8. Pair & Peer Interview Practice
9. Practising HR, Behavioural and Situational Questions
10. Handling HR Questions (Extempore)
11. Company Specific & Profile Specific Interviews (AI Tools)
12. Online Spoken English Test

Total Laboratory Hours: 30 hours

Mode of Evaluation:

Evaluation Scheme			
MSE	CA	ESE	Total Marks
-	50	-	50
50			

Course Code: CS317P	Course Name: Problem Solving Lab-2	L	T	P	C
		0	0	2	1

Pre-requisite: Knowledge of Data Structure

Course Objectives:

1. To familiarize students with the basic and advance of Problem Solving.
2. To learn modern techniques to solve DSA Problems.

Course Outcomes: After completion of the lab, the student will be able to

1.Apply data-driven and combinatorial problem-solving techniques
2.Design solutions using Dynamic Programming
3.Develop solutions using trees and graph algorithms
4. Integrate advanced techniques to solve complex problems

CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)

CO -PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	1	2	-	-	-	-	-	1
CO2	3	3	3	1	2	-	-	-	-	-	1
CO3	3	3	3	2	2	-	-	-	-	-	1
CO4	2	3	3	2	3	-	-	2	1	1	2

List of Experiments:

1. Hashing Basics- 1. Frequency counting 2. First non-repeating character 3. Two sum using hashing
2. Hash Map + Sets Advanced -1. Group anagrams 2. Longest consecutive sequence 3. Subarray sum = k
3. Prefix Sum / Difference Array - 1. Range sum queries 2. Rainwater trapping 3.Kadane's algorithm
4. Recursion +Backtracking Advanced- 1. Sudoku solver 2. Word search in grid 3. Combination sum
5. Dynamic Programming– I (Introduction) - 1 . Fibonacci DP 2. Climbing stairs 3. Min cost climbing stairs
6. DP – II (1D DP) 1. House robber 2. Decode ways 3. Partition equal subset sum
7. DP – III (2D DP) 1. Longest common subsequence 2. Edit distance 3. 0/1 knapsack
8. DP – IV (Advanced) 1. Matrix chain multiplication 2. Palindromic subsequence 3. LIS
9. Trees Basics(Conceptual Only)- 1. Preorder traversal 2. Height of tree 3. BFS on tree
10. Binary Search Tree Techniques - 1. Validate BST 2. LCA 3. Kth smallest element
11. Graphs – BFS/DFS 1. Number of islands 2. Bipartite graph check 3. Connected components
12. Graphs – Shortest Path 1. Dijkstra 2. Bellman–Ford 3. Shortest path in grid
13. Graphs – Advanced 1 . Topological sort 2. Course schedule 3. Detect cycle in directed graph
14. Mixed Techniques / DSA Integration - 1. Sliding window + hashing problem 2. Graph + BFS+ DP hybrid 3. String + DP complex case
15. Final Mini Project 1. 20-question problem sheet 2. Contest style evaluation 3. Full pattern-based lab exam

Total Laboratory Hours: 30 hours

Mode of Evaluation:

Evaluation Scheme			
MSE	**CA**	**ESE**	**Total Marks**
-	**50**	-	**50**
50			